

REGIONAL CLOSED FORMS FOR $\zeta(s)$: AN $\text{SO}(4)$ PROJECTION AND MACHINE-CHECKED REFORMULATIONS OF RH

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ABSTRACT. Position. This note records a machine-checked geometric interpretation of the Riemann zeta function in the public HQIV Lean library [5], building on the $\mathfrak{so}(8)$ closure record [4], the octonion light-cone carrier [3], and the TUFT+SM synthesis [1]. It is a *formal mathematics* paper, not a phenomenology forecast: every theorem is a regional closed form, projection identity, or logical equivalence stated in standard analytic notation and verified by `lake build HQIVStory`.

Main geometric claim. The functional-equation reflection $\sigma \leftrightarrow 1 - \sigma$ is realized as a 45° rotation in a quaternionic plane. The fixed diagonal component is $1/\sqrt{2}$; the variable *equator factor* $(2\sigma - 1)/\sqrt{2}$ (Lean: `so4CriticalFactor`) is the sin-cos carrier on the strip. Classical $\zeta(s)$ is then a *piecewise* closed form on $\mathbb{C}$ in the usual sense: Dirichlet/Euler for $\Re s > 1$, the functional-equation sin-cos- Γ assembly on the open strip, Bernoulli values at negative integers, and π -power formulas at even positives.

Open-strip assembly. The functional-equation expression for $\zeta(s)$ on $0 < \Re s < 1$ combines $\Gamma(1-s)$ poles, oscillating sin/cos factors, and apparent singularities that obscure where the *nontrivial* zeros can live until the regional bookkeeping is separated (§7): trivial zeros and Γ poles are classified explicitly; the equator factor vanishes only on $\Re s = \frac{1}{2}$; and the meaningful zero question is pinned to that critical line. Pairwise cancellation along the FE orbit $(s, 1-s)$ holds for every σ , but only pointwise vanishing of the equator factor forces $\sigma = \frac{1}{2}$ (§8).

Sine/cosine trichotomy. At even integers $s = 2k$ the angle $\pi s/2 = k\pi$ kills the sine factor ($\sin = 0$, $\cos = (-1)^k$): even ζ -values are pure π -powers. At odd integers the cosine factor vanishes and sine carries the residue (no elementary closed form for $\zeta(3), \zeta(5), \dots$). On the fractional open strip both factors are nonzero and $\tan(\pi s/2)$ is the rotation ratio.

RH and Goldbach. Classically RH and Goldbach are independent open problems. Here both are tied to the same normalized slope $\frac{1}{2}$: critical-line localization for nontrivial ζ -zeros, and midpoint slope $N/(p+q) = \frac{1}{2}$ for Goldbach prime pairs (§12). Lean proves an *exact equivalence* between a single bridge hypothesis (`S08ProjectedHalfSlopeBridge 2`) and `RiemannHypothesis` $\wedge$ `GoldbachParity`—not a one-sided implication with extra conditions. Proving the bridge remains as hard as the conjunction of the two classical conjectures.

Status. Every algebraic / Mathlib link in the harmonic-prime- ζ chain, every 45° projection identity, the $\Re s$ -dependence lemmas, the FE quadruplet orbit formulation, the bridge equivalence $\text{RH} \wedge \text{Goldbach} \Leftrightarrow \text{half-slope bridge}$, the discrete-dominates-continuum heat-flow comparison with exact phase lock, the concrete- Λ port of the de Bruijn–Newman constant, the tangent orbit law, the spectral-cancellation profile of zeros (no cancellation in the Euler product region; resonance blow-up of $1/\zeta$ at every zero), and the resonance-channeling reduction (two-channel dichotomy; `InteriorAssemblyNonzero` $\Leftrightarrow$ RH; RH $\Leftrightarrow$ square-root spectral weights at every zero), the vertical-axis rigidity (the zero set admits at most one mirror axis, necessarily $\Re s = \frac{1}{2}$), the Diophantine transformer laws (trace ladder $\rightarrow \zeta$; FE composition = $\text{diag}(1/n)$; adjoint law $\Leftrightarrow$ line; RH $\Leftrightarrow$ adjoint law at every zero), and the octonionic associator channel (commutative saturation of the transformer family; non-associative torsion provably active at every zero; RH $\Leftrightarrow$ torsion weights), and the $S^7/\text{SO}(8)$ torsion cancellation laws (single S^7 point; chirality flip; RH $\Leftrightarrow$ FE torsion cancellation at every zero), and the log edge (polar decoupling; prime-log incommensurability; two prime phases pin the height; the Goldbach circle $pq + (q - N)^2 = N^2$) cited here are sorry-free in `lake build HQIVStory`. Unconditional *Riemann Hypothesis* and *Goldbach parity* are *not* claimed: the main conditional hypothesis `InteriorAssemblyNonzeroAtNontrivialZerosOffLine` and a witness for the (provably equivalent) half-slope bridge remain open (Appendix A).

Scope (formal mathematics, not phenomenology). This note records machine-checked regional closed forms, projection identities, and logical equivalences in the public HQIV Lean library. It does *not* assert phenomenological predictions, a scale witness, or patch-closed completeness in the TUFT+SM sense. When classical notation appears (Γ , $\sin/\cos$, heat flow, de Bruijn–Newman families), it serves as a *comparison dictionary* for readers trained in analytic number theory; the proved objects are discrete harmonic sums, shell weights, prime Euler products, and explicit predicates on the open strip. No fundamental continuum is required for the theorems cited here; a literal continuum limit on the load-bearing shell index would erase the harmonic– Δ closure that motivates the construction ([4]).

1. INTRODUCTION

The Riemann zeta function is usually introduced as a Dirichlet series and then analytically continued. On the open critical strip $0 < \Re s < 1$, the continued formula is the standard $\sin$ – $\cos$ – Γ functional-equation assembly: products of Γ , π , $\sin(\pi s/2)$, and $\cos(\pi s/2)$ that look over-determined and singular until trivial zeros and continuation slots are separated (§7). This note gives a geometric interpretation of that assembly: the functional equation is reflection symmetry $\sigma \leftrightarrow 1 - \sigma$ rotated by 45° ; a fixed diagonal component separates from a variable *equator factor* $(2\sigma - 1)/\sqrt{2}$; and the divergent harmonic series of the closure paper forces a phase-lift connector Δ that closes $\mathfrak{so}(3) + \Delta$ to $\mathfrak{so}(4)$ on a four-dimensional quaternion carrier.

This note records what is already proved in Lean:

- (i) the harmonic– Δ – $\text{SO}(4)$ closure bridge;
- (ii) the harmonic $\Rightarrow$ prime Euler product $\Rightarrow \zeta \Rightarrow \Lambda = -\zeta'/\zeta$ pipeline for $\Re s > 1$;
- (iii) regional closed forms for $\zeta(s)$ matching Mathlib;
- (iv) sine/cosine factor cancellation at even π -multiples and rotation ratios on odd / fractional arguments;
- (v) exclusion lemmas ruling out nontrivial zeros in half-planes and at negative integer slots;
- (vi) a factorization $\zeta(s) = h(s) \cdot (2\sigma - 1)/\sqrt{2}$ on the open strip off $\sigma = \frac{1}{2}$, with $h = \text{interiorStripH}$;
- (vii) dependence on $\Re s$ only: the equator factor is a function of $\sigma = \Re s$ and cannot globally identify ζ with the projection formula (§3.1);
- (viii) FE + Schwarz quadruplet orbits for ζ -zeros, with unconditional FE pairs and conditional full quadruplets (§8.1);
- (ix) the shared $\frac{1}{2}$ slope bridge: exact equivalence of $\text{RH} \wedge \text{Goldbach}$ with a single geometric hypothesis (§12);
- (x) the discrete/continuum heat-flow comparison $c\|H_t\| \leq \|\text{HQIVDeformedSum}\|$ with exact phase alignment (§11.5), and the de Bruijn–Newman constant ported as a concrete Lean definition with Rodgers–Tao and Newman/dBN as named statements (§11.6);
- (xi) the tangent law of the quadruplet orbit: FE inverts and Schwarz conjugates $T(s) = \tan(\pi s/2)$, and the critical line is exactly the unimodular-tangent locus (§8.2);
- (xii) zeros as spectral cancellations of the Euler product: no cancellation in the product region, breakdown-locus confinement, and resonance blow-up of $\|1/\zeta\|$ at every zero (§8.3);
- (xiii) resonance channeling: the two-channel dichotomy at every zero, the equivalence $\text{InteriorAssemblyNonzero} \Leftrightarrow \text{RH}$, the $\text{SO}(4)$ spectral-line transformer $n \mapsto n^{-s}$ with FE pairing $\|n^{-s}\| \|n^{-(1-s)}\| = 1/n$, and $\text{RH} \Leftrightarrow$ square-root spectral weights at every zero (§8.4);
- (xiv) vertical axis rigidity: composing the FE point-reflection with any second mirror axis yields a translation that marches zeros out of the strip, so the zero set admits at most one vertical axis and it must be $\Re s = \frac{1}{2}$ (§8.5);

- (xv) same-height orbit collapse: the quadruplet partner $1 - \bar{\rho}$ shares ρ 's height and merges with it exactly on the line, giving $\text{RH} \Leftrightarrow$ “one zero per height” — unconditionally in the $|\mathfrak{S}|$ form, and with explicit Schwarz hypothesis in the strict form (§8.6);
- (xvi) polar projection collapse: the polar separation $1 - \bar{s} - s = -\sqrt{2} \cdot \text{so4CriticalFactor}(s)$ is the equator factor itself, and single-point polar projection on the zero set is RH (§8.7);
- (xvii) the zero–holonomy–Goldbach chain: every zero activates every midpoint pair, each pair carries Hopf holonomy support, and the half-slope bridge rewrites as polar collapse $\wedge$ parity (§8.8);
- (xviii) the TUFT nested frame tower: level- N spectral frames on odd spheres S^{2N-1} , the harmonic radius law $\text{frameNormSq} = H_N \Leftrightarrow \Re s = \frac{1}{2}$ at every level, divergence of the radius ladder on the line, and $\text{RH} \Leftrightarrow$ harmonic radii at every zero (§8.9);
- (xix) the Diophantine transformer: the level- N diagonal operator $D_N(s) = \text{diag}(1^{-s}, \dots, N^{-s})$ whose trace ladder converges to ζ , whose Hilbert–Schmidt norm is the tower radius, whose FE composition is the fixed rational harmonic operator $\text{diag}(1/n)$, and whose adjoint law $D_N(1-s) = D_N(s)^\dagger \Leftrightarrow \Re s = \frac{1}{2}$ gives $\text{RH} \Leftrightarrow$ self-adjointness across FE at every zero (§8.10);
- (xx) the octonionic associator channel: the transformer family is provably commutative and normal (the formal reason the diagonal attack saturates), while the non-associative carrier escapes — the triple (e_1, e_2, e_4) has $[e_1, e_2, e_4] = -e_3 - e_4 \neq 0$ on the frozen Fano tables, prime spectral weights ride the associator trilinearly, the channel $2(abc)^{-2\sigma}$ never vanishes (zeros kill the abelian assembly, never the torsion), and $\text{RH} \Leftrightarrow$ square-root torsion weight at every zero (§8.11);
- (xxi) S^7 torsion cancellation under $\text{SO}(8)$: the torsion vector is a positive multiple of one fixed carrier direction (the S^7 shadow never moves), the FE defect is odd with a chirality flip across the line, FE partners' torsions share an $\text{SO}(8)$ orbit iff they are equal iff $\Re s = \frac{1}{2}$, and $\text{RH} \Leftrightarrow$ the torsion defect vanishes at every zero (§8.12);
- (xxii) the critical-line carrier bundle: the strip embeds as a cylinder $\sigma \times S^1$, but only at $\Re s = \frac{1}{2}$ is the lift the full Hopf circle ($r = 1, f = 0$, rolling map); off the line the fiber shrinks ($r < 1$), the tangent sorts the half-planes hyperbolically, and quadruplet collapse fails—whole-strip classification is conditional (§5);
- (xxiii) the log edge: polar decoupling $n^{-s} = n^{-\sigma} \cdot e^{-it \log n}$ (modulus knows only σ , phase only t), Diophantine independence of prime phase speeds ($k \log p = m \log q \Rightarrow k = m = 0$, from unique factorization), two prime phases pin the height exactly, the Goldbach circle $pq + (q - N)^2 = N^2$ as literal additive curvature with phase-speed cap $\log(pq) \leq 2 \log N$, and — under RH — two prime phases as a complete address for a zero (§8.13).

The remaining RH step is *not* continuation; it is proving that the assembly h does not vanish at nontrivial zeros off the line—equivalently, upgrading orbit cancellation (functional-equation pairs) to pointwise cancellation at actual ζ zeros. Goldbach is the same geometric half-slope on the additive prime channel, equivalent only under the holonomy bridge hypothesis.

1.1. Why zeta functions appear in HEP and other HQIV integrations. HQIV does not use “zeta” as a single object. Three layers recur across the library and paper series; this note isolates the *classical projection* layer and explains why the same vocabulary appears in high-energy phenomenology readouts without implying that every “zeta” is Riemann’s function.

Sector zeta on the patch carrier. In the TUFT+SM synthesis [1], *sector zeta* means fibre-torsion subleading spectral bodies on nested Hopf shells: Beltrami-sector determinants, **zetaHQIVTerm** phase on the null-lattice carrier, and holonomy discharge bundles on $\mathfrak{so}(8)$. These are *finite* shell-indexed readouts tied to mass rungs and coupling ladders on the accessible patch—not the complex-analytic $\zeta(s)$ studied here.

Classical $\zeta(s)$ as projection geometry (this note). The present manuscript shows that Mathlib’s Riemann zeta is already a *regional closed form* of an $\text{SO}(4)$ projection: harmonic shell weights

sum to $\zeta(s)$ for $\Re s > 1$; prime Euler products and $\Lambda = -\zeta'/\zeta$ follow on the same region; the open strip is a sin-cos- Γ assembly with a 45° equator factor $(2\sigma - 1)/\sqrt{2}$. The proved polar law $n^{-s} = n^{-\sigma} \cdot e^{-it \log n}$ (§8.13) is the structural reason modulus-based equivalent conditions throughout this note reformulate RH without importing phase data into σ .

HEP multichannel and decay formulas. The heavy-flavour decay programme [2] uses the same *spectral-log* pattern at the formula layer: discrete multichannel branching rules, three-ledger β tipping, and rational EM contact coefficients are computed from composite traces and shell weights without partial-width inputs. Zeta language enters because (i) prime and shell indices carry multiplicative weights n^{-s} (or their real-axis cousins) in every channel; (ii) logarithmic phase speeds $\log p$, $\log N$ tie additive midpoint data (Goldbach circle, §8.13) to multiplicative pooling (quarkonium channels, joint products pq); and (iii) continuation and resonance (zeros of assembled factors, blow-up of $1/\zeta$, spectral cancellation profile, §8.3) is the same geometric template as “channel factors vanish where the global assembly vanishes.” This note does not reproduce HEP numerics; it supplies the proved analytic spine that makes the shared vocabulary precise: zeta functions in HQIV integrations are projection formulas for prime/shell spectral data, not ad hoc naming.

Continuum notation elsewhere. Heat-flow, de Bruijn–Newman, and O-Maxwell continuum charts in sibling manuscripts are *comparison dictionaries* for classical literature—as in `readout_dictionary_messaging.tex` above, not load-bearing continuum extensions. The discrete harmonic ladder and shell index remain the proved ontology; a literal continuum limit on those indices would erase the predictions ([4]).

2. HARMONIC DIVERGENCE FORCES Δ AND $\text{SO}(4)$

The closure spine [4] proves a uniform lower bound

$$H_n := \sum_{m=0}^{n-1} \frac{1}{m+1} \leq K(n) := \sum_{m=0}^{n-1} \underbrace{\frac{1}{m+1} (1 + \alpha \log(m+1))}_{\text{shell density}}, \quad \alpha = \frac{3}{5},$$

with K strictly increasing ([Hqiv/Story/S3ClosureDeltaLiftBridge.lean](#), [Hqiv/Geometry/OctonionicLightCone.lean](#)). The decomposition

$$K(n) = H_n + \alpha \sum_{m=0}^{n-1} \frac{\log(m+1)}{m+1}$$

is the exact curvature-integral split (`curvature\_integral\_eq\_harmonic\_plus\_alpha\_log`).

Δ is *not* identical to H_n , but it is forced because the normalized channel $\Omega = K/K_*$ diverges. On the toy four-dimensional carrier the Lie span of $\mathfrak{so}(3)$ together with the connector $\Delta_4 = J_{14}$ generates full $\mathfrak{so}(4)$ (`so3\_delta\_lifts\_to\_so4`).

2.1. Third harmonic orbit and the $1 + \alpha/3$ projection. The third partial sum $H_3 = 1 + \frac{1}{2} + \frac{1}{3}$ is the first shell carrying an explicit $1/3$ harmonic term. Against the HQIV imprint $\alpha = 3/5$ the projection multiplier

$$1 + \frac{\alpha}{3} = 1 + \frac{1}{5} = \frac{6}{5} = 1.2$$

is proved exactly ([Hqiv/Story/S3HarmonicDeltaEvenOrbit.lean](#), `so4HarmonicThirdProjection\_eq\_one\_point\_two`). On the eight-dimensional shell the unit 7-sphere area proxy is $\pi^4/3$; halving at the 45° equator gives $\pi^4/6$ (`so4EquatorHalfArea\_eq\_pi\_four\_sixths`), parallel to the even Bernoulli slot $\zeta(2) = \pi^2/6$ at π^2 power.

3. FORTY-FIVE DEGREE PROJECTION

Definition 3.1 (Functional-equation pair and 45° slots). *For $\sigma \in \mathbb{R}$ define the pair $(\sigma, 1 - \sigma)$ and the rotated coordinates*

$$\begin{aligned} \text{rot45Diag}(\sigma, 1 - \sigma) &= \frac{\sigma + (1 - \sigma)}{\sqrt{2}} = \frac{1}{\sqrt{2}}, \\ \text{rot45Free}(\sigma, 1 - \sigma) &= \frac{\sigma - (1 - \sigma)}{\sqrt{2}} = \frac{2\sigma - 1}{\sqrt{2}}. \end{aligned}$$

At $\theta = \pi/4$ one has $\cos \theta = \sin \theta = \sqrt{2}/2$ (not π); π enters through the functional-equation angle $\pi s/2$ below.

Theorem 3.2 (Critical line as equator). *The free coordinate vanishes iff $\sigma = \frac{1}{2}$:*

$$\text{rot45Free}(\sigma, 1 - \sigma) = 0 \iff \sigma = \frac{1}{2}.$$

A nonzero rotation offset $\pi/4 + \delta$ with $\sin \delta \neq 0$ moves the vanishing locus off $\sigma = \frac{1}{2}$ ([Hqiv/Story/S3RotationRigidity.lean](#)).

The complex lift of the equator factor is the **SO(4) critical factor**

$$\text{so4CriticalFactor}(s) = \frac{2\sigma - 1}{\sqrt{2}}, \quad s = \sigma + it,$$

which vanishes exactly on $\Re s = \frac{1}{2}$ ([Hqiv/Story/S3ZetaClosedForm.lean](#)).

3.1. The equator factor depends only on $\Re s$. The exact 45° projection is distinguished: other rotation angles move the vanishing locus off $\sigma = \frac{1}{2}$ ([Hqiv/Story/S3RotationRigidity.lean](#)). What the projection formula *cannot* decide is also proved:

- **σ -blindness.** `exactTwiddleReadout` and `so4CriticalFactor` depend only on $\sigma = \Re s$, not on height $t = \Im s$ ([Hqiv/Story/S3SigmaReadoutScope.lean](#), `sigmaReadout\_blind\_to\_imaginary\_height`).
- **Identification $\nRightarrow$ constraint.** Pointwise equality $\zeta(s) = \text{exactTwiddleReadout}(s)$ transfers zero iff at s , but does not force $\sigma = \frac{1}{2}$ without that identification (`zeta\_eq\_readout\_same\_zero\_set`).
- **No global identification.** $\zeta(s) = \text{exactTwiddleReadout}(s)$ for all s fails: $\zeta(0) = -\frac{1}{2}$ but the projection formula gives $-1/\sqrt{2}$ (`not\_globally\_zeta\_eq\_real\_twiddle\_readout`).
- **Affine form.** Off the even lift, $\text{so4CriticalFactor}(z) = (2\Re z - 1)/\sqrt{2}$ (`so4CriticalFactor\_eq\_affine\_re`); this is real-valued in σ and not holomorphic in s .

Proving regional ζ closed forms therefore *relocates* zeros onto the construction without transferring the equator vanishing locus onto ζ .

3.2. Mirrored twiddle pairs. The functional equation supplies reflection $\sigma \leftrightarrow 1 - \sigma$. Rather than treating twiddle angles as independent knobs, one may pair angles that are mirrors under the same reflection:

$$\theta \longleftrightarrow \frac{\pi}{2} - \theta.$$

[Hqiv/Story/S3MirroredTwiddlePair.lean](#) formalizes two complementary readouts.

Rotation twiddle (paper default). On the FE pair $(\sigma, 1 - \sigma)$ the free coordinate after rotation by θ is

$$\text{rotFree}_\theta(\sigma) = \sigma(\cos \theta + \sin \theta) - \sin \theta,$$

with vanishing locus the **projection line** $\sigma_0(\theta) = \sin \theta / (\cos \theta + \sin \theta)$ ([Hqiv/Story/S3RotationRigidity.lean](#)). Its mirror at $\pi/2 - \theta$ has zero locus

$$\sigma_0\left(\frac{\pi}{2} - \theta\right) = 1 - \sigma_0(\theta),$$

so the two candidate axes are symmetric about $\sigma = \frac{1}{2}$ (**projectionLine_mirror**). Across the FE orbit the pair cancels pairwise:

$$\text{rotFree}_\theta(\sigma, 1 - \sigma) + \text{rotFree}_{\pi/2 - \theta}(1 - \sigma, \sigma) = 0$$

(**rotFree_mirror_pair_sum**); on a fixed pair $(\sigma, 1 - \sigma)$ the sum and difference separate into

$$(\cos \theta + \sin \theta)(2\sigma - 1) \quad \text{and} \quad \cos \theta - \sin \theta,$$

so the critical line is the **coincidence locus** where the two channels agree in the σ -only sense: the difference is constant in σ and vanishes only at $\theta = \pi/4$, when both angles coincide and $\sigma_0(\pi/4) = \frac{1}{2}$. Affine twiddle (weighted σ form). The weighted combination

$$\text{affineTwiddle}_\theta(\sigma) = \cos \theta \sigma + \sin \theta (1 - \sigma)$$

is the mirror swap $\sin \theta \sigma + \cos \theta (1 - \sigma)$ at $\pi/2 - \theta$ (**affineTwiddle_mirror_angle**). When $\sin \theta \neq \cos \theta$ its zero is $\sigma = \sin \theta / (\sin \theta - \cos \theta)$, again mirrored by $\sigma \mapsto 1 - \sigma$ (**affineTwiddleZero_mirror**). At $\theta = \pi/4$ this family is *degenerate*: $\cos \theta = \sin \theta$ so the readout is the constant $\sin(\pi/4) \neq 0$ and never vanishes (**affineTwiddle_pi_div_four_ne_zero**); the 45° critical-line pin is native to the rotation/normalized form, not this affine parameterization.

Example: 30° and 60°. $\pi/6$ and $\pi/3$ are a mirror pair. Their affine zero loci are far from $\sigma = \frac{1}{2}$ (e.g. at $\pi/6$ the zero is $(1/2)/(1/2 - \sqrt{3}/2)$), but they satisfy the exact swap $\sigma_0(\pi/3) = 1 - \sigma_0(\pi/6)$ (**affineTwiddleZero_pi_six_pi_three_mirror**). Near-45° rotation angles instead give projection lines $\sigma = \frac{1}{2} \pm \varepsilon$ — the regime where both mirrors hug the critical line.

Interaction with sine/cosine slots and vertical rigidity. The FE channel already carries the angle $\pi s/2$ in $\sin(\pi s/2)$ and $\cos(\pi s/2)$ (§4); mirrored twiddles add a second angular degree on the real σ pair, enriching the even/odd/fractional projection trichotomy without replacing it. Each individual twiddle still defines a straight vertical candidate axis; if two such mirrors are imposed together with FE closure, the same translation argument as in §8.5 forces any common strip zero locus onto the unique fixed line $\Re s = \frac{1}{2}$. Off the line the pair distinguishes *orbit* cancellation (FE partners) from *pointwise* coincidence on $\sigma = \frac{1}{2}$ — a two-channel sharpening of the orbit-vs-pointwise gap (§8).

3.3. Higher twiddles as probes of the factorization engine. The linear-twiddle family and its cubic specializations are **not** intended to yield new information about the location of zeros on their own loci. Their purpose is to illuminate the **factorization engine** — the decomposition of $\zeta(s)$ into an interior assembly times a geometric readout factor — when the readout is taken from channels other than the distinguished 45° projection.

Main channel (paper default). Off the critical line on the open strip,

$$\zeta(s) = h(s) \cdot \text{so4CriticalFactor}(s) = h(s) \cdot \frac{2\sigma - 1}{\sqrt{2}}$$

(§9, (2)): the interior assembly $h = \text{interiorStripH}$ absorbs everything that is not the equator readout.

Higher-twiddle readouts. For a coefficient $c \in (0, 1)$ the linear free readout is

$$\text{free}_c(\sigma) = c \cdot \sigma + (1 - c) \cdot (1 - \sigma),$$

with zero on the vertical line $\sigma_c = (c - 1)/(2c - 1)$ when $2c - 1 \neq 0$. Functional-equation reflection interchanges c with $1 - c$, producing mirrored pairs (§3.2). Specializing to the plastic constant via $c_\rho = 1 - \rho^{-1}$ (unique $\rho > 1$ with $\rho^3 = \rho + 1$) produces the normalized cubic twiddle `so4PlasticCubicFree` whose zero locus is the explicit algebraic number

$$\sigma_\rho = \frac{1}{2 - \rho}$$

([Hqiv/Story/S3CubicPlasticTwiddle.lean](#): mirror symmetry, normalization, and FE partner `so4PlasticCubicFree\_one\_sub`; patch factorization off the probe locus in [Hqiv/Story/S3HigherTwiddleFactorizationProbe.lean](#)). This locus lies far from $\Re s = \frac{1}{2}$; the construction is a *controlled perturbation* of the readout geometry, not a competing critical-line candidate.

Multi-channel laboratory. Because cubic (and, more generally, higher-degree) twiddles are not expected to constrain zeros directly on their own axes, they serve as probes of **cross-channel** behavior inside the factorization architecture:

- How does the interior assembly change when the geometric factor is `so4PlasticCubicFree` (or another `free_c`) instead of `so4CriticalFactor`?
- What is the coupling between the main 45° channel and a higher-twiddle channel when both are measured at the same strip point?
- Can one define a controlled cross-talk term supported only on semiprime spectral weights?

The patch-level target, not yet formalized, is a family

$$\zeta(s) = h_c(s) \cdot \text{free}_c(\Re s) \quad (\text{off the zero locus of channel } c),$$

parallel to (2) but with readout channel c replacing the 45° equator factor.

Semiprime-mediated influence on the critical line. Any influence a higher-twiddle channel exerts on points lying on the 45° critical line is expected to be **semiprime mediated**: it can reach the main channel only through spectral weights supported on integers with exactly two prime factors (semiprimes pq with $p \neq q$ in the square-free case). This mirrors the arithmetic difficulty of locating or controlling semiprimes in the additive Goldbach channel (§12): midpoint pairs $p + q = 2N$ are easy to *certify* once found, while products pq are the composite objects that factorization-side bookkeeping must resolve ([Hqiv/Geometry/GoldbachG2Parity.lean](#), composite-branch and G_2 -triple decomposition lemmas). Consequently higher twiddles may affect the factorization engine at critical-line points only *indirectly and weakly* — not as new zero locators, but as cross-talk that must pass through the hardest multiplicative spectral lines.

Open technical tasks.

- Define $h_c(s)$ (or h_ρ) relative to a higher-twiddle readout, matching the interior assembly role of `interiorStripH`.
- Prove the corresponding patch-level factorization inside finite spectral frames of level N .
- Make precise the semiprime support of any cross-talk between the 45° channel and higher-twiddle channels (a leakage term vanishing on primes and higher-composite weights). Lean already types finite-frame leakage with `SemiprimeSupportedWeight` and proves prime spectral lines carry zero weight ([Hqiv/Story/S3HigherTwiddleFactorizationProbe.lean](#), `semiprime\_weight\_kills\_prime\_spectral\_line`).

4. QUATERNION AXIS PROJECTION

On the unit imaginary quaternion shell $S^3 \subset \mathbb{H}$ the six axis poles $\pm i, \pm j, \pm k$ project to $\pm 1/\sqrt{2}$ on single axes and cancel as antipodal orbit pairs ([Hqiv/Story/S3SixPolesResidual.lean](#), [Hqiv/Story/S3PrimeAxisCancellation.lean](#)).

Definition 4.1 (Sine and cosine factors in the FE channel). *Define the sine and cosine factors*

$$\mathcal{S}(s) = \sin \frac{\pi s}{2}, \quad \mathcal{C}(s) = \cos \frac{\pi s}{2}.$$

These are the oscillating factors in the functional-equation odd channel; $\mathcal{C}$ is `zetaSinCosFactor` in Lean.

Theorem 4.2 (Even π -multiples: sine vanishes). *For $k \in \mathbb{N}$ at $s = 2k$:*

$$\mathcal{S}(2k) = \sin(k\pi) = 0, \quad \mathcal{C}(2k) = \cos(k\pi) = (-1)^k.$$

Theorem 4.3 (Odd integers: cosine cancels). *For $k \in \mathbb{N}$ at $s = 2k + 1$:*

$$\mathcal{C}(2k + 1) = 0, \quad \mathcal{S}(2k + 1) = (-1)^k.$$

Theorem 4.4 (Fractional strip: rotation ratio). *For $0 < \Re s < 1$ with $\mathcal{C}(s) \neq 0$,*

$$\frac{\mathcal{S}(s)}{\mathcal{C}(s)} = \tan \frac{\pi s}{2}.$$

At $\sigma = \frac{1}{2}$ both slots equal $\sqrt{2}/2$ and the ratio is 1 (`zetaAxisRotationRatio\_at\_half`).

Theorems 4.2–4.4 are proved in [Hqiv/Story/S3ZetaAxisRotationProjection.lean](#). They explain why **even** ζ -values admit π -sector closed forms while **odd** positive integers (e.g. $\zeta(3)$) occupy the Apéry / plastic continuation slot with no elementary closed form in Mathlib.

5. ANALYTIC STRIP LIFT: CYLINDER $\sigma \times S^1$ IN $\mathbb{H}$

The critical strip $0 < \sigma < 1$ embeds as a *cylinder cover*: at fixed σ the height $t = \Im s$ rolls a Hopf fiber on $S^3 \subset \mathbb{H}$, while σ enters through the 45° free coordinate of §3. The geometric punch is not “the whole strip is a flat cylinder” but **what is special about the critical line**: there the free coordinate vanishes, the fiber radius is maximal, the lift is the rolling map, and the tangent readout is unimodular with quadruplet collapse; off the line the analytic slots carry hyperbolic ($\cosh / \sinh$) sorting and the Hopf circle is strictly shrunk. Lean packages the contrast in [Hqiv/Story/S3CriticalLineCarrierBundle.lean](#); the lift and conditional whole-strip classification remain in [Hqiv/Story/S3AnalyticStripLift.lean](#) and [Hqiv/Story/S3AnalyticStripClassification.lean](#).

5.1. Lift and Hopf fiber.

Definition 5.1 (Analytic strip lift). *For $\sigma, t \in \mathbb{R}$ define the **free coordinate***

$$f(\sigma) = \frac{2\sigma - 1}{\sqrt{2}} = \text{rot45Free}(\sigma, 1 - \sigma),$$

*the **Hopf fiber radius** $r(\sigma) = \sqrt{1 - f(\sigma)^2}$, and the lift*

$$\text{stripAnalyticLift}(\sigma, t) = (0, f(\sigma), r(\sigma) \cos t, r(\sigma) \sin t)$$

on the $p_0 = 0$ slice of S^3 . For $s = \sigma + it$ set $\text{stripPointLift}(s) = \text{stripAnalyticLift}(\sigma, t)$.

Theorem 5.2 (On-shell and σ -equator). *For $0 \leq \sigma \leq 1$ and all t , the lift lies on S^3 (`strip\_analytic\_lift\_on\_s3`). The free coordinate vanishes exactly on the critical line: $f(\sigma) = 0 \Leftrightarrow \sigma = \frac{1}{2}$ (`strip\_sigma\_free\_coord\_vanishes\_iff`). At $\sigma = \frac{1}{2}$ the lift agrees with the rolling map of [Hqiv/Story/S3StripRollingProjection.lean](#): $\text{stripAnalyticLift}(\frac{1}{2}, t) = \text{stripRollingMap}(t)$ (`strip\_analytic\_lift\_eq\_rolling`).*

5.2. Critical line carrier bundle.

Theorem 5.3 (Line is the unique maximal-circle, unimodular locus). *On the open strip $0 < \sigma < 1$, the following are equivalent (`critical\_line\_carrier\_iff\_chain`):*

$$\Re s = \frac{1}{2} \iff f(\sigma) = 0 \wedge r(\sigma) = 1 \wedge \text{criticalLineDeviation}(s) = 0 \wedge \text{so4CriticalFactor}(s) = 0 \wedge \|\tan(\pi s/2)\|$$

Here $f(\sigma) = \text{stripSigmaFreeCoord}(\sigma)$ and $r(\sigma) = \text{stripHopfFiberRadius}(\sigma)$. The lift on the line is the rolling map (`critical\_line\_lift\_is\_rolling`).

Theorem 5.4 (Off-line carrier tax). *For $0 \leq \sigma \leq 1$ with $\sigma \neq \frac{1}{2}$, $f(\sigma) \neq 0$ and $r(\sigma) < 1$ (`off\_line\_carrier\_shrinks\_hopf\_fiber`, `strip\_hopf\_fiber\_radius\_lt\_one\_iff`). On the open strip with $\Re s \neq \frac{1}{2}$, the tangent is not unimodular, the quadruplet does not collapse, and σ sorts into the disk or exterior of the unit circle (`off\_line\_hyperbolic\_tangent\_sorts`, `critical\_strip\_off\_line\_quadruplet\_does\_not\_collapse`). Cancellation off the line requires the full offset $f(\sigma) + r(\sigma)(\cos t + \sin t) = 0$, not merely $\cos t + \sin t = 0$ (`strip\_analytic\_cancellation\_off\_line\_needs\_free`).*

Theorem 5.5 (Path C on the line via rolling). *Under `StripAnalyticLiftMatches` and `ZetaEqualsS3ResidualAt`, for $\Re s = \frac{1}{2}$,*

$$\zeta(s) = 0 \iff \text{ZeroProducingOrbit}(\text{stripRollingMap}(t)) \wedge \text{PathEStripHolonomyCloses}(s)$$

(`critical\_line\_pathC\_classification\_via\_rolling`); equivalently, only the rolling orbit is needed on the line (`critical\_line\_pathC\_rolling\_only`).

The hyperbolic character off-line is in the *analytic* $\sin/\cos$ slots ($\cosh/\sinh$ factors cancel only in $\|\sin\|^2 - \|\cos\|^2 = -\cos(\pi\sigma)$, `normSq\_proj\_sin\_sub\_cos`), not in the constraint $f(\sigma)^2 + r(\sigma)^2 = 1$ (a unit circle in the lift chart, `strip\_lift\_free\_radius\_unit\_circle`).

5.3. $\mathbb{C}$ cover versus $\mathbb{H}$ circle. In $\mathbb{C}$ the height t is a *non-compact* cover coordinate. On the lift, t and $t + 2\pi n$ define the same point on S^3 (`strip\_analytic\_lift\_period`, `sameStripHeight`). The Hopf-base trace $(r(\sigma) \cos t, r(\sigma) \sin t)$ is likewise 2π -periodic in t (`strip\_hopf\_circle\_period`). Thus each fixed σ carries an S^1 fiber inside $\mathbb{H}$ while $\mathbb{C}$ records the universal cover — the geometric form of “countably infinite heights, compact phase carrier” discussed in the spectral literature without the quaternion chart.

5.4. Harmonic-shell arc sweep (not heights t). Counting by scanning $t \in \mathbb{R}$ conflates the universal cover with a counter. On the line, $t \neq t + 2\pi$ yet the same fiber class is recovered (`strip\_height\_not\_injective\_on\_circle`, `strip\_rolling\_map\_respects\_height\_cover`).

The promoted counting axis pairs harmonic depth H_n with angular *arc partitions*. At shell $n > 0$ the Hopf circle is cut into n arcs of width $2\pi/n$, indexed by k at polar angle $2\pi k/n$ (`shellSweepAngle`, `shellArcWidth`). Finer shells refine the sweep when $n \mid m$ (`sweep\_refines\_coarser\_angle`, `sweep\_arc\_width\_refines`). Slots form a countable union of finite partitions: `HarmonicShellSlot` = $\Sigma_n \text{Fin } n$ injects into $\mathbb{N}$ (`encodeHarmonicShellSlot\_injective`).

Monotone ticks $1, 2, 3, \dots$ order shells by cumulative weight H_n only. Sweeping the circle adds *spatial resolution on S^1* : shell n is an n -way partition of the same compact carrier, not one more real height. Divergent H_n supplies depth; $2\pi k/n$ supplies angular resolution (`harmonic\_depth\_diverges`, `HarmonicSweepCountingAxis`). Each slot carries primitive-root phase on S^1 (`shell\_slot\_phase\_eq\_exp\_angle`) and rolls to `stripRollingMap(2\pi k/n)` (`strip\_rolling\_at\_slot\_eq\_lift`). Head/tail reflection is antipodal, $\theta \mapsto \theta + \pi$ (`head\_tail\_reflect\_rolling\_antipodal`); reflections quotient slots, they do not enumerate them.

Reciprocal j - k chart. At shell n define depth reciprocal $j_n = 1/(\pi H_n)$ and angular slot $k_n = k/n$. The j - k phase exponent is $\pi j_n k_n = k/(n H_n)$ (`harmonicShellJKExponent`, `harmonic\_shell\_jk\_exponent\_eq`). The critical-line probe point is $s = \frac{1}{2} + i k/(n H_n)$ (`criticalLinePointAtHarmonicSlot`); the unit-circle phase is $e^{i\pi j_n k_n}$ (`harmonicShellJKPhase`). **Unconditional** balance is `criticalProj(stripRollingMap( $\pi j_n k_n$ )) = 0` (`harmonicShellSlotIsBalance`); **conditional** ζ -zero is $\zeta(s) = 0$ under `RollingZetaIdentificationAtCriticalLine` (`harmonicShellSlotIsZetaZ`). **Honesty.** $\zeta(s) = 0$ on the line as cover balance still requires `RollingZetaIdentificationAtCriticalLine` (`zeta\_zero\_iff\_cover\_balance`); the counting axis (n, k) is unconditional (`harmonicShellZeroCountingBun`).

5.5. 45° readout on the whole strip. The 45° projection amplitude on the lift is

$$\text{criticalProj}(\text{stripAnalyticLift}(\sigma, t)) = \frac{f(\sigma) + r(\sigma)(\cos t + \sin t)}{\sqrt{2}}$$

(`strip\_analytic\_critical\_proj`). Cancellation is the balanced condition $f(\sigma) + r(\sigma)(\cos t + \sin t) = 0$ (`strip\_analytic\_cancellation\_iff`). The Fourier twiddle `stripAnalyticTwiddle( $\sigma, t$ ) =  $e^{it}$  · criticalProj( $\dots$ )` extends the critical-line rolling twiddle to all σ (`strip\_analytic\_twiddle\_on\_line`).

5.6. Conditional zero classification. Honesty. As in §7, identifying $\zeta(s)$ with the lifted residual requires `ZetaEqualsS3ResidualAt` and the sample hypothesis `StripAnalyticLiftMatches` (`P.coords = stripPointLift(s)`).

Theorem 5.6 (Whole-strip cancellation form). *Under `StripAnalyticLiftMatches` and `ZetaEqualsS3ResidualAt`,*

$$\zeta(s) = 0 \iff f(\sigma) + r(\sigma)(\cos t + \sin t) = 0$$

on the open strip (`zeta\_zero\_iff\_strip\_analytic\_cancellation`).

Theorem 5.7 (Path C/E on the cylinder). *Under the same hypotheses, for $0 < \sigma < 1$,*

$$\zeta(s) = 0 \iff \text{ZeroProducingOrbit}(\text{stripPointLift}(s)) \wedge \text{PathEStripHolonomyCloses}(s)$$

(`analytic\_strip\_pathC\_classification`). On $\sigma = \frac{1}{2}$ this recovers the rolling classification of [Hqiv/Story/S3StripRollingProjection.lean](#) (`analytic\_strip\_rolling\_classification\_on\_line`).

The capstone `InteriorAssemblyNonzeroAtNontrivialZerosOffLine` is unchanged: the lift supplies *geometry* on $\sigma \times S^1$; the identification bridge remains RH-hard.

6. REGIONAL CLOSED FORM FOR $\zeta(s)$

Table 1 collects the proved regional formulas. Every row is machine-checked against Mathlib in [Hqiv/Story/S3ZetaClosedForm.lean](#) and [Hqiv/Story/S3S04ZetaProjectionClosedForm.lean](#).

6.1. Right half-plane.

Theorem 6.1 (Shell / Dirichlet sum). *For $\Re s > 1$,*

$$\zeta(s) = \sum_{n=0}^{\infty} (n+1)^{-s},$$

the HQIV shell ladder with shift $(n+1)$ (`riemannZeta\_dirichlet\_closed\_form`).

This is the even harmonic–Euler sector: the harmonic–prime– ζ path proves the prime product and $\Lambda = -\zeta'/\zeta$ on the same region ([Hqiv/Story/S3HarmonicPrimeZetaPath.lean](#)).

TABLE 1. Regional closed forms for $\zeta(s)$ from the $\text{SO}(4)$ projection (all rows proved).

Region	Channel	Closed form
$\Re s > 1$	even / Dirichlet	$\sum_{n \geq 0} (n+1)^{-s}$
$0 < \Re s < 1$	odd / sin-cos- Γ	FE assembly (<code>oddStripChannel</code>)
$s = 2k, k \geq 1$	even π -sector	$(-1)^{k+1} \frac{2^{2k-1} \pi^{2k} B_{2k}}{(2k)!}$
$s = -k$	Bernoulli	$-\frac{B'_{k+1}}{k+1}$
odd $s \geq 3$	odd continuation	no elementary closed form

6.2. Open critical strip.

Theorem 6.2 (Functional-equation closed form). *For $0 < \Re s < 1$,*

$$(1) \quad \zeta(s) = 2(2\pi)^{-(1-s)} \Gamma(1-s) \cos \frac{\pi(1-s)}{2} \zeta(1-s).$$

Equation (1) is Mathlib's `riemannZeta_one_sub`, repackaged as `oddStripChannel =  $\zeta(s)$` on the strip ([Hqiv/Story/S3S04InteriorWitness.lean](#)).

6.3. Even positive integers.

Theorem 6.3 (π -sector values). *For $k \geq 1$,*

$$\zeta(2k) = (-1)^{k+1} \frac{2^{2k-1} \pi^{2k} B_{2k}}{(2k)!}.$$

In particular $\zeta(2) = \pi^2/6$ and $\zeta(4) = \pi^4/90$.

At these points $\mathcal{S}(2k) = 0$: the sine/cosine rotation has completed full π -turns and only the even π -sector residue remains—consistent with Theorem 4.2.

6.4. Negative integers.

Theorem 6.4 (Bernoulli slot). *For $k \in \mathbb{N}$,*

$$\zeta(-k) = -\frac{B'_{k+1}}{k+1}.$$

6.5. Master projection formula.

Definition 6.5. $\text{zetaSO4Projected}(s) := \zeta(s)$ with the regional characterisations of Table 1 as proved corollaries.

Off the critical line on the open strip,

$$\zeta(s) = \text{so4EquatorProject}(s) \cdot \text{interiorStripH}(s), \quad \text{so4EquatorProject}(s) = \frac{2\sigma - 1}{\sqrt{2}},$$

when $0 < \sigma < 1$ and $\sigma \neq \frac{1}{2}$ (`zetaSO4Projected\_eq\_sin\_cos\_channel\_off\_line`).

7. OPEN-STRIP ASSEMBLY AND THE CRITICAL LINE

The open-strip functional-equation assembly (1) exposes three features that look singular when read naively:

- (a) $\Gamma(1-s)$ has poles at $s = 1, 2, 3, \dots$;
- (b) $\cos(\pi(1-s)/2)$ has zeros at odd integers;
- (c) negative-even *trivial* zeros of ζ sit at $s = -2, -4, \dots$

The machine-checked classification treats these as **regional bookkeeping**, not as contradictions:

- **Negative-even trivial zeros** are the only negative-integer zeros; nontrivial zeros cannot occupy negative integer slots ([Hqiv/Story/S3FESlotDischarge.lean](#), `nontrivial\_zero\_fe\_slot`).
- **Open strip** points avoid $\Re s > 1$ and $\Re s < 0$ by Mathlib nonvanishing and FE reflection ([Hqiv/Story/S3DeltaOrbitOffStrip.lean](#)).
- Γ **poles** on the strip are paired with $\cos$ zeros in the standard FE; they mark the even-sector continuation, not a complex off-line zero locus.

8. ORBIT CANCELLATION VS. POINTWISE VANISHING

[Hqiv/Story/S3OrbitVsPointwiseGap.lean](#) separates three layers:

- (i) **Orbit cancellation.** For every σ , $\text{rot45Free}(\sigma) + \text{rot45Free}(1-\sigma) = 0$ (`orbit\_free\_sum\_cancels`); this holds even off $\Re s = \frac{1}{2}$ and cannot exclude off-line ζ zeros by itself.
- (ii) **Pointwise vanishing.** A single free coordinate vanishes iff $\sigma = \frac{1}{2}$ (`pointwise\_free\_zero\_iff\_on\_line`).
- (iii) **RH equivalence.** $\text{AllNontrivialZerosOnLine} \Leftrightarrow \text{RiemannHypothesis}$ (`allNontrivialZerosOnLine\_iff\_RiemannHypothesis`).

8.1. FE + Schwarz quadruplet. Nontrivial zeros satisfy the FE pair $\zeta(1-s) = 0$ unconditionally ([Hqiv/Story/S3ZeroQuadrupletOrbit.lean](#), `zeta\_zero\_fe\_pair`). The full Schwarz quadruplet

$$\{s, 1-s, \bar{s}, 1-\bar{s}\}$$

is packaged as `zetaZeroQuadruplet`; all four points carry zeros when the conjugation identity $\zeta(\bar{t}) = \overline{\zeta(t)}$ is available (`zeta\_zero\_quadruplet`). On $\Re s = \frac{1}{2}$ the reflections $\bar{s}$ and $1-s$ coincide, collapsing the quadruplet to an FE pair—the same collapse as orbit vs. pointwise on the σ channel.

Remark 8.1 (Geometric reading). *The “infinity” from Γ and the “zero” from negative-even slots are two faces of the same projection geometry: they clear the plane of spurious complex-axis candidates and leave the pointwise equator condition $\text{rot45Free} = 0 \Leftrightarrow \sigma = \frac{1}{2}$ as the only balanced σ -only dependence. RH asks that actual ζ zeros satisfy this stronger pointwise condition, not merely orbit cancellation along $\sigma \leftrightarrow 1-\sigma$.*

8.2. Tangent law of the orbit: factorization angles and the unimodular locus. The strip closed form carries a *factorization angle* $\pi s/2$: the sine/cosine rotation slots are $\sin(\pi s/2)$ and $\cos(\pi s/2)$ and their ratio $T(s) = \tan(\pi s/2)$ is the projection formula (`zetaAxisRotationRatio`). [Hqiv/Story/S3TangentOrbitCriticalLine.lean](#) proves the orbit law for that angle under the FE + Schwarz quadruplet — all of it unconditional projection geometry, no RH input. The orbit acts on the tangent by inversion and conjugation. FE reflection swaps the slots exactly ($\sin(\pi(1-s)/2) = \cos(\pi s/2)$, `proj\_sin\_one\_sub`), so

$$T(1-s) = T(s)^{-1}, \quad T(\bar{s}) = \overline{T(s)}, \quad T(1-\bar{s}) = \overline{T(s)}^{-1}$$

(`projTangent\_one\_sub`, `projTangent\_schwarz`, `projTangent\_quadruplet`), with the orbit reciprocity $T(1-s)T(s) = 1$ on the strip (`projTangent\_orbit\_reciprocity`). The quadruplet

$\{s, 1-s, \bar{s}, 1-\bar{s}\}$ reads out through the tangent as $\{T, T^{-1}, \bar{T}, \bar{T}^{-1}\}$: the group $\{1, \text{inv}, \text{conj}, \text{inv} \circ \text{conj}\}$.

The critical line is exactly the unimodular locus. The engine is the exact modulus identity

$$\|\sin(\pi s/2)\|^2 - \|\cos(\pi s/2)\|^2 = -\cos(\pi \Re s)$$

(`normSq\proj\sin\sub\cos`): the $\Im s$ content rides in common cosh / sinh factors that cancel in the difference — the σ -blindness of §3.1, reappearing at the tangent level. Consequently, on the open strip,

$$\|T(s)\|^2 = 1 \iff \Re s = \frac{1}{2}, \quad \|T(s)\|^2 < 1 \iff \Re s < \frac{1}{2}, \quad \|T(s)\|^2 > 1 \iff \Re s > \frac{1}{2}$$

(`normSq\projTangent\eq\one\iff` and companions): the tangent modulus is a *complete invariant* for the critical line and sorts the strip around it.

Orbit collapse $\Leftrightarrow$ critical line. Inversion equals conjugation on the tangent — $T^{-1} = \bar{T}$, collapsing the quadruplet tangent orbit from four values to two — precisely on the critical line (`projTangent\inv\eq\conj\iff`). And the tie to the ζ factorization is exact: the equator factor of $\zeta = h \cdot (2\sigma - 1) / \sqrt{2}$ vanishes exactly where the tangent is unimodular,

$$\text{so4CriticalFactor}(s) = 0 \iff \|\tan(\pi s/2)\|^2 = 1$$

(`so4CriticalFactor\zero\iff\unimodular\tangent`). The critical line is thus characterized three ways at once — vanishing equator factor, unimodular factorization tangent, and coincidence of the two orbit symmetries — all proved to be the same locus.

8.3. Zeros as spectral cancellations of the Euler product. The reading that drives this entire construction — *the zeros are exactly the spectral cancellations that give the Euler prime product* — has three provable faces, and [Hqiv/Story/S3EulerSpectralCancellation.lean](#) proves all three. None of them requires RH.

In the product region the prime phases provably cannot cancel. For $\Re s > 1$ every Euler factor is individually nonvanishing, $\|p^{-s}\| = p^{-\Re s} < 1$ so $(1 - p^{-s})^{-1} \neq 0$ (`euler\factor\ne\zero`); the product of these phases converges to $\zeta(s)$ and the value is nonzero (`euler\product\region\no\_cancellation`, packaging Mathlib's `riemannZeta.eulerProduct_hasProd` with the nonvanishing). Where the spectral product is a genuine convergent product, cancellation can never *complete*: a zero would contradict the product representation itself.

Zeros live exactly on the breakdown locus. Contrapositively, $\zeta(s) = 0$ forces $\Re s < 1$ — impossible anywhere the product region extends, including the closed boundary $\Re s = 1$ (`zero\forces\_product\_breakdown`). Combined with the strip confinement of §7, every nontrivial zero sits in the open strip $0 < \Re s < 1$ (`spectral\_cancellation\_in\_breakdown\_locus`) — precisely the region where the prime spectral series no longer converges absolutely. The zero set lives on the breakdown locus of the Euler product and nowhere else.

Each zero is a resonance pole of the inverse spectral response. On the product region the prime spectrum is *tame*: the von Mangoldt series converges and equals the log-derivative, $\sum_n \Lambda(n)n^{-s} = -\zeta'(s)/\zeta(s)$ (`spectral\_series\_tame`). The *inverse* spectral response $1/\zeta$ — the un-inverted prime-phase product $\prod_p (1 - p^{-s})$ continued to the strip — blows up at every zero:

$$\|\zeta(s)^{-1}\| \longrightarrow \infty \quad \text{as } s \rightarrow \rho \text{ along the nonvanishing locus,}$$

unconditionally for any zero $\rho \neq 1$ (`inverse\_spectral\_response\_blows\_up\_at\_zero`); the punctured-neighbourhood version carries the classical isolated-zero fact as an explicit hypothesis rather than smuggling it in (`inverse\_spectral\_response\_blows\_up\_punctured`). A completed cancellation is a *resonance*: the spectral measure of the primes diverges exactly there.

Capstone: the spectral-cancellation profile. `spectral\_cancellation\_profile` packages, for any nontrivial zero ρ : (i) open-strip confinement (breakdown region); (ii) impossibility in the product region; (iii) the resonance blow-up of $\|1/\zeta\|$ at ρ ; and (iv) the quadruplet tangent orbit $\{T, T^{-1}, \bar{T}, \bar{T}^{-1}\}$ of §8.2 attached to the factorization angle $\pi\rho/2$. A nontrivial zero *is* a completed spectral cancellation carrying the full orbit data. What RH adds — and only RH — is that every such cancellation sits on the unimodular-tangent locus $\|T\| = 1$.

8.4. Resonance channeling and the $\text{SO}(4)$ spectral-line transformer. [Hqiv/Story/S3SpectralResonanceChanneling.lean](#) combines the three proved layers — tangency, equator factor, and spectral cancellation — into the sharpest forcing statements available without assuming RH.

The cancellation is collective. A single prime line never cancels on the entire right half-plane: for $\Re s > 0$ (in particular on the whole strip), $1 - p^{-s} \neq 0$ (`euler\_factor\_base\_ne\_zero\_right\_half`), so every finite spectral truncation $\prod_{p \in F} (1 - p^{-s})$ is nonzero (`finite\_spectral\_truncation\_ne\_zero`). A zeta zero needs all primes at once; no finite interference produces it.

The two-channel dichotomy. On the strip $\zeta = h \cdot (2\sigma - 1)/\sqrt{2}$ off the line, so at any nontrivial zero the collective cancellation must be absorbed by exactly one of two channels (`resonance\_channeling\_dichotomy`):

$$\underbrace{\Re \rho = \frac{1}{2} : \text{so4CriticalFactor}(\rho) = 0, \|T(\rho)\|^2 = 1}_{\text{geometric equator channel absorbs}} \quad \text{or} \quad \underbrace{\Re \rho \neq \frac{1}{2} : \text{interiorStripH}(\rho) = 0}_{\text{free assembly forced to vanish}}$$

(`offline\_zero\_forces\_assembly\_vanish`). RH is exactly the statement that the second channel never fires, and this is now a proved *equivalence*, not merely an implication: `geometric\_channel\_absorption\_iff\_RH` gives `InteriorAssemblyNonzeroAtNontrivialZerosOffLine` `interiorStrip` RH.

The $\text{SO}(4)$ spectral-line transformer. For a given n , the transformer is the spectral line `so4SpectralLine` $n s = n^{-s}$, with proved laws: the modulus law $\|n^{-s}\| = n^{-\Re s}$ (σ -blindness at the spectrum level, `so4SpectralLine\_norm`); the unconditional *FE pairing*

$$\|n^{-s}\| \cdot \|n^{-(1-s)}\| = \frac{1}{n} \quad \text{for all } s$$

(`so4SpectralLine\_fe\_product`); the *geometric-mean condition* — the two paired lines are equal iff $\Re s = \frac{1}{2}$ (`so4SpectralLine\_geometric\_mean\_iff`); the *square-root-weight condition* $\|n^{-s}\|^2 = 1/n \Leftrightarrow \Re s = \frac{1}{2}$ (`so4SpectralLine\_sq\_weight`); multiplicativity $(ab)^{-s} = a^{-s}b^{-s}$ (`so4SpectralLine\_mul`); and the prime generation of every line — every $n \geq 2$ is a multiple of a prime and its line factors through that prime's line (`spectral\_line\_prime\_factorization`). Main conditional hypothesis. `RH\_iff\_zero\_spectral\_weights`: **RH is equivalent to the statement that at every nontrivial zero, every spectral line carries exactly square-root weight** $\|n^{-\rho}\|^2 = 1/n$. “For every prime, the zero sits at its square-root pole weight” — made exact and proved RH-equivalent. And `zero\_line\_characterizations` shows the three equivalent conditions agree at every zero: vanishing equator factor $\Leftrightarrow$ unimodular tangent $\Leftrightarrow$ all lines at square-root weight $\Leftrightarrow \Re \rho = \frac{1}{2}$. The unconditional content is a genuine *reduction*: RH is now a single-channel statement — the free assembly h never absorbs a collective cancellation — with every geometric equivalent condition proved to name the same line.

8.5. Vertical axis rigidity: the orbits cannot wobble. [Hqiv/Story/S3VerticalAxisRigidity.lean](#) formalizes the rigidity intuition: any “curvature” of the zero locus — a second symmetry axis, a tilted or bent critical line — would force zeros *out of the strip, off to infinity*.

Two mirrors make a translation. The zero set is unconditionally closed under the FE point-reflection $\rho \mapsto 1 - \rho$ (`nontrivial\_zero\_fe\_closed` — FE only, no Schwarz hypothesis needed). Suppose it were *also* closed under the mirror `verticalReflect` $\sigma_0 : s \mapsto 2\sigma_0 - \bar{s}$ about some axis $\Re s = \sigma_0$. Composing the two symmetries twice yields the translation

$$s \mapsto s + (2 - 4\sigma_0),$$

nontrivial unless $\sigma_0 = \frac{1}{2}$; iterating it marches any zero monotonically in $\Re$, straight out of the open strip — contradicting the proved strip confinement. Hence (`strip\_set\_unique\_vertical\_axis`, abstract; `zeta\_zero\_unique\_vertical\_axis`, instantiated): **the nontrivial zero set admits at most one vertical mirror axis, and it must be $\Re s = \frac{1}{2}$** . The only hypothesis carried by the zeta instantiation is nonemptiness of the zero set (Hardy 1914, classical, not yet in Mathlib).

Supporting rigidity facts (all unconditional). The fixed-point set of any vertical mirror is exactly its axis — a straight line, never a curve (`verticalReflect\_fixed\_iff`); the mirror about $\frac{1}{2}$ is the FE+Schwarz composite $1 - \bar{s}$ (`verticalReflect\_half\_eq\_one\_sub\_schwarz`). If a point and its FE partner $1 - s$ share a vertical line, that line is $\Re s = \frac{1}{2}$ (`fe\_pair\_same\_vertical\_line\_iff`): any two symmetry-partner points on one line force *the* line. And the candidate locus itself has no curvature anywhere: as a subset of all of $\mathbb{C}$, the square-root-weight locus is exactly the straight vertical line $\{\Re s = \frac{1}{2}\}$ (`sq\_weight\_locus\_is\_vertical\_line`), swept by the vertical flow $s \mapsto s + it$ with no transverse drift (`sq\_weight\_locus\_vertical\_flow`).

Honest scope. “Two points on the line force all points onto the line” is not a valid inference classically. What is proved: (i) two symmetry-partner points on a common vertical line pin that line to $\frac{1}{2}$; (ii) the candidate locus is a perfect straight line, not a curve; (iii) the zero set cannot support a second mirror axis without leaking out of the strip. Promoting “the unique possible axis is $\frac{1}{2}$ ” to “every zero lies on it” is exactly RH.

8.6. Same-height orbit collapse: “one zero per height” is exactly RH. [Hqiv/Story/S3SameHeightOrbitCollapse.lean](#) scopes the two-part equivalence — *collectivity pins the axis; two zeros at the same height are off the board* — and extracts the genuine theorem inside it.

Collectivity does not pin the axis. The cancellation at every zero is provably collective, but at *any* point of the strip the all-primes weight family is the perfectly consistent $\|n^{-s}\|^2 = n^{-2\sigma}$ (`spectral\_weights\_consistent\_at\_any\_sigma`): nothing diverges or contradicts off the line. What distinguishes $\sigma = \frac{1}{2}$ is solely the square-root normalization n^{-1} — and that normalization *is* RH (`RH\_iff\_zero\_spectral\_weights`).

The same-height partner and the merge law. The quadruplet member $1 - \bar{s}$ sits at the *same height*, $\Im(1 - \bar{s}) = \Im s$ (`one\_sub\_schwarz\_same\_height`), and merges with s exactly on the critical line (`same\_height\_partner\_merges\_iff` — the mirror fixed-point law again). An off-line zero would therefore carry a *distinct* second zero at its own height; an on-line zero carries none.

“One zero per height” $\Leftrightarrow$ RH.. Declaring the straddling case off the board is not a proof of RH — it is an exact reformulation of RH, proved as an equivalence in two strengths:

- `RH\_iff\_height\_pins\_re` (**FE only, unconditional**): $\text{RH} \Leftrightarrow$ any two nontrivial zeros with the same $|\Im|$ have the same $\Re$. The backward direction needs no Schwarz input: the FE partner $1 - \rho$ already supplies a zero of equal $|\Im|$ with reflected real part, forcing $\Re \rho = 1 - \Re \rho$.
- `RH\_iff\_unique\_zero\_per\_height` (Schwarz identity carried as the explicit hypothesis, in the style of `zeta\_zero\_quadruplet`): $\text{RH} \Leftrightarrow$ each height carries at most one nontrivial zero.

The orbit picture is correct: a quadruplet either straddles the line with two distinct same-height members or collapses onto it. Both directions are machine-checked with exact equivalence.

8.7. Polar projection collapse: the poles must land on one point. [Hqiv/Story/S3PolarProjectionCollapse.lean](#) formalizes the polar reading: *the poles of S^3 must project to a*

single point on the complex plane for the zero to sit on the line — and resolves the worry that “SO(4) is only applicable if RH is true.”

The polar separation is the equator factor. At the height of s the mirror presents two pole images, s and $\text{polarPartner } s = 1 - \bar{s}$, always at the same height ($\text{polarPartner_same_height}$). Their separation is horizontal, real, and *exactly* the equator factor:

$$\text{polarPartner } s - s = 1 - 2\sigma = -\sqrt{2} \cdot \text{so4CriticalFactor}(s)$$

($\text{polar_separation_eq, polar_separation_eq_twiddle}$), with norm equal to twice the distance to the critical line ($\text{polar_separation_eq_twice_deviation}$). The 45° equator factor is, verbatim, the displacement between the two pole projections: off the line the SO(4) geometry does not break — it *reports* the polar separation through the surviving equator channel, which is precisely the off-line branch of the channeling dichotomy of §8.4.

Single-point projection $\Leftrightarrow$ RH.. The two pole images coincide iff $\Re s = \frac{1}{2}$ ($\text{polar_collapse_iff_on_line}$). Single-point polar projection on the zero set ($\text{PolarProjectionCollapsesOnZeros}$) is therefore not a hidden presupposition of the SO(4) machinery but a well-formed proposition, and it is RH — proved as a exact-equivalence equivalence ($\text{polar_collapse_iff_RH}$), chaining to “one zero per height” given the Schwarz identity ($\text{polar_collapse_iff_unique_height}$). Single-point polar projection at every zero is thus the fourth face of RH in this stack, equal in strength to unimodular tangent, square-root spectral weights, and one-zero-per-height.

8.8. Zero–holonomy–Goldbach chain: every zero activates every pair. [Hqiv/Story/S3ZeroHolonomyGoldbachChain.lean](#) chains the collectivity of the cancellation through the Goldbach parity bridge: since all zeros are comprised of all primes, every zero carries the holonomy of every Goldbach pair.

Activation and the AM–GM weight cap (unconditional). At any nontrivial zero ρ and for any midpoint pair $p+q = 2N$, both prime lines are nonvanishing participants in the collective cancellation ($\text{zero_activates_pair}$), and the joint line factors multiplicatively through the pair. AM–GM gives $pq \leq N^2$ ($\text{midpoint_pair_product_le}$), so on the critical line the joint pair weight obeys

$$\|(pq)^{-s}\|^2 = \frac{1}{pq} \geq \frac{1}{N^2}$$

($\text{midpoint_pair_joint_weight_bound}$): the *additive* half-slope datum N caps the *multiplicative* spectral weight of its pairs.

Every zero contains the holonomy of every pair. The main chain theorem $\text{zero_contains_pair_holonomy}$ states, at any nontrivial zero and for any midpoint pair: activation of both prime lines, multiplicative factorization of the joint line, SO(4) midpoint slope $\frac{1}{2}$, and the Hopf-fiber holonomy support certificate (integrable shell with $\text{S08AdmissibleHolonomy } \Delta/G_2/\text{triality fields}$) — all unconditional. Under GoldbachParity , every midpoint $N \geq 2$ admits a holonomy-supported pair ($\text{parity_gives_holonomy_support}$).

The bridge in geometric-channel language. The SO(8) half-slope bridge now rewrites with both channels geometric:

$$\begin{aligned} \text{S08ProjectedHalfSlopeBridge } 2 &\iff (\text{square-root spectral weights at every zero}) \wedge \text{parity} \\ &\iff (\text{single-point polar projection at every zero}) \wedge \text{parity} \end{aligned}$$

($\text{bridge_iff_spectral_weights_and_parity, bridge_iff_polar_collapse_and_parity}$). The chain does not prove either side — the bridge remains exactly RH $\wedge$ Goldbach with exact equivalence — but both sides now speak the same SO(4)/holonomy dialect: zeta-side polar collapse, prime-side holonomy-supported midpoint pairs.

8.9. TUFT nested frame tower: odd spheres and the harmonic radius ladder. [Hqiv/Story/S3TuftNestedFrameTower.lean](#) formalizes what the TUFT nested fibration — odd spheres S^{2n+1} for every n — buys. At level N the first N spectral lines assemble into the frame vector $(1^{-s}, \dots, N^{-s}) \in \mathbb{C}^N$, whose normalized version lives on S^{2N-1} ; the structure is carried by the squared radius $\text{spectralFrameNormSq } N s = \sum_{n \leq N} \|n^{-s}\|^2$.

Tower structure (unconditional). The levels nest with the new line's weight as increment (`spectralFrameNormSq\_succ`); every level is σ -blind, so the vertical flow moves the frame on a sphere of *fixed* radius (`spectralFrameNormSq\_eq\_of\_re\_eq`); the radius never degenerates (`spectralFrameNormSq\_pos`).

The harmonic radius law. The rigidity payoff is a *ladder of equivalent conditions*: for every level $N \geq 2$,

$$\text{spectralFrameNormSq } N s = H_N \iff \Re s = \frac{1}{2}$$

(`frame\_radius\_eq\_harmonic\_iff`), where H_N is the harmonic backbone of the closure story. The critical line is the unique vertical line on which the nested sphere radii reproduce the harmonic ladder — *the divergent series that forced Δ and closed $\mathfrak{so}(3) + \Delta$ to $\mathfrak{so}(4)$ in the first place*. The loop closes: on the line the tower rebuilds the harmonic divergence (`frame\_tower\_diverges\_on\_line`, via Mathlib's divergence of the harmonic series), and the closure inequality $H_N \leq K(N)$ reads as a curvature bound on the nested sphere radii (`frame\_radius\_le\_curvature\_on\_line`). What it buys — in classical terms. `RH\_iff\_zero\_frame\_tower\_harmonic`: $\mathrm{RH} \Leftrightarrow$ at every nontrivial zero the *whole tower* carries harmonic radii — an infinite ladder of equivalent reformulations, one per level, all collapsing to the same bit. The tower adds consistency rigidity (infinitely many independent conditions provably naming one line, with the closure backbone as reference values), not a new forcing mechanism: each level is RH -equivalent, not RH -implying.

8.10. The Diophantine transformer: the operator behind the tower. The tower, the spectral lines, and the FE pairing are all shadows of one concrete object. [Hqiv/Story/S3DiophantineTransformer.lean](#) defines the level- N *Diophantine transformer*

$$D_N(s) = \text{diag}(1^{-s}, 2^{-s}, \dots, N^{-s}) \in M_N(\mathbb{C}),$$

an integer-indexed, prime-generated diagonal operator on $\mathbb{C}^N$ (entries are multiplicative in the index, so the primes generate everything), and proves its laws unconditionally.

Trace ladder $\rightarrow \zeta$. $\text{tr } D_N(s)$ is the N -th Dirichlet partial sum (`diophantineTransformer\_trace`), and on the product region $\Re s > 1$ the trace ladder converges: $\text{tr } D_N(s) \rightarrow \zeta(s)$ (`diophantineTransformer\_trace\_tendsto`). The zeta function *is* the trace of the transformer tower.

HS norm = tower radius. The squared Hilbert–Schmidt norm of $D_N(s)$ is exactly $\text{spectralFrameNormSq } N s$ (`diophantineTransformer\_hs\_normSq`): the TUFT frame radius of §8.9 is the operator's norm ladder, and the harmonic radius law becomes “ D_N is harmonically normalized ($\|D_N\|_{\text{HS}}^2 = H_N$) exactly on the critical line.”

FE composition is a fixed rational operator. For *every* s ,

$$D_N(s) \cdot D_N(1-s) = \text{diag}(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{N})$$

(`diophantineTransformer\_fe\_product`) — the s -dependence cancels completely and what remains is a pure Diophantine object with rational entries, whose trace is the harmonic backbone H_N (`feProduct\_trace\_eq\_harmonic`). The functional equation, seen through the transformer, is the statement that the two reflected operators always compose to the *integer harmonic operator*.

The adjoint law names the line. The genuinely new rigidity is at the level of adjoints: for any $n \geq 2$,

$$n^{-(1-s)} = \overline{n^{-s}} \iff \Re s = \frac{1}{2}$$

(`transformer\_entry\_adjoint\_iff`), hence for $N \geq 2$

$$D_N(1-s) = D_N(s)^\dagger \iff \Re s = \frac{1}{2}$$

(`diophantineTransformer\_adjoint\_iff`). The FE reflection agrees with the operator adjoint precisely on the critical line, and there the transformer is a *scaled isometry* whose unitarity defect is again the fixed harmonic diagonal: $D_N(s)D_N(s)^\dagger = \text{diag}(1/n)$ (`transformer\_scaled\_isometry\_on\_line`).

RH in operator language. `RH\_iff\_transformer\_adjoint\_at\_zeros`: $\text{RH} \Leftrightarrow$ at every nontrivial zero and every level $N \geq 2$, $D_N(1-\rho) = D_N(\rho)^\dagger$. This is the machine-checked form of the Hilbert–Pólya intuition the construction has been circling: not “there exists a self-adjoint operator whose spectrum is the zeros,” but “ $\text{RH} \Leftrightarrow$ the concrete transformer family is self-adjoint-across-FE at the zeros.” As with every equivalent condition in this paper, the law is an exact reformulation of RH, not a proof of RH.

8.11. The octonionic associator channel: breaking the abelian obstruction. Diagonalization itself is the obstruction. [Hqiv/Story/S30octonionicAssociatorChannel.lean](#) first proves the saturation formally: the transformer family is *abelian* and *normal*,

$$D_N(s) D_N(s') = D_N(s') D_N(s), \quad D_N(s) D_N(s)^\dagger = D_N(s)^\dagger D_N(s)$$

(`diophantineTransformer\_mul\_comm`, `diophantineTransformer\_normal`). A commuting normal family is simultaneously diagonalizable; every invariant it carries factors through the entries, and the entries are exactly the σ -blind / RH-equivalent readouts catalogued above. This is the machine-checked “good reason” the diagonal attack cannot go further.

The escape hatch is non-associativity. The HQIV stack carries one algebra that no commuting family can simulate: the octonionic light-cone carrier, whose associator $[x, y, z] = (xy)z - x(yz)$ is the same phase channel that supplies CP-odd structure and Lorentz bookkeeping elsewhere in the framework. On the frozen Fano tables the witness is concrete:

$$[e_1, e_2, e_4] = (e_1 e_2) e_4 - e_1 (e_2 e_4) = e_7 e_4 - e_1 (-e_5) = -e_3 - e_4 \neq 0,$$

with squared norm 2 (`octonionAssociator\_e1\_e2\_e4`, `octonionAssociatorNormSq\_e1\_e2\_e4`).

Spectral weights ride the torsion. The associator is trilinear in scalar weights (`octonionAssociator\_smul`), so the prime spectral moduli couple cleanly: weighting (e_1, e_2, e_4) by $\|a^{-s}\|, \|b^{-s}\|, \|c^{-s}\|$ defines the *associator channel* with closed form

$$\text{octAssociatorChannel } a b c s = 2(abc)^{-2\sigma}$$

(`octAssociatorChannel\_eq`). Three unconditional laws follow:

- **Zeros cannot cancel the torsion** (`zero\_keeps\_associator\_torsion`): the channel is strictly positive at every point — in particular at every nontrivial zero. The zero annihilates the abelian assembly $\sum n^{-s}$; the non-associative channel survives untouched, carrying every prime’s weight through the cancellation.
- **Asymmetry condition** (`octAssociatorChannel\_eq\_iff`): the channel carries the square-root weight $2/(abc)$ exactly on $\Re s = \frac{1}{2}$.
- **FE pairing** (`octAssociatorChannel\_fe\_product`): $\text{channel}(s) \cdot \text{channel}(1-s) = 4/(abc)^2$ for every s — the associator analogue of $D_N(s)D_N(1-s) = \text{diag}(1/n)$.

The Goldbach hook also extends: for a midpoint pair $p + q = 2N$, the triple $(p, q, 2N)$ keeps the channel above $1/N^3$ on the line (`midpoint\_triple\_channel\_floor`) — the AM–GM cap pushed into the non-associative channel.

What the asymmetry buys — in classical terms. `RH\_iff\_zero\_associator\_channel`: $\text{RH} \Leftrightarrow$ at every nontrivial zero the channel carries exact square-root weight for every triple. The channel structurally breaks the abelian obstruction — it is an orientation-sensitive three-slot invariant that no commuting diagonal family can produce, and unlike the assembly it is provably *active* at every zero. But its modulus coupling is still driven by $\Re s$ alone, so the condition collapses to the same single bit: another exact reformulation. What is genuinely new is the *shape* of the invariant (a non-associative three-form versus abelian diagonals) and the unconditional positivity at zeros; converting the torsion’s phase content — rather than its modulus — into a forcing mechanism is the open edge this section names.

8.12. S^7 torsion cancellation: the $\text{SO}(8)$ twist. [Hqiv/Story/S3OctonionS7TorsionCancellation.lean](#) keeps the full *torsion vector* `octTorsionVector a b c s` $\in \mathbb{R}^8$ in the carrier and asks where it cancels between FE partners. The carrier’s unit sphere is S^7 ; the rotation group is $\text{SO}(8)$, whose complete invariant on $\mathbb{R}^8$ is the norm.

The S^7 shadow never moves. For every spectral parameter,

$$\text{octTorsionVector } a b c s = (abc)^{-\sigma} \cdot (-e_3 - e_4)$$

`(octTorsionVector\_eq)`: a *positive* radial weight times one fixed carrier direction `(octTorsion\_single\_point\_on\_s7)`. The entire spectral flow is radial; S^7 sees a single point, and all the information sits in the weight $(abc)^{-\sigma}$.

Cancellation, chirality, orbits. The FE defect `octTorsionDefect = torsion(s) - torsion(1 - s)` is the weight gap $(abc)^{-\sigma} - (abc)^{-(1-\sigma)}$ times the fixed direction `(octTorsionDefect\_eq)`, is odd under $s \mapsto 1 - s$ `(octTorsionDefect\_antisymm)`, and:

- **cancels exactly on the line** `(octTorsionDefect\_zero\_iff)`: $\text{torsion}(s) = \text{torsion}(1 - s) \Leftrightarrow \Re s = \frac{1}{2}$;
- **flips chirality across the line** `(octTorsionWeightGap\_pos\_iff, octTorsionWeightGap\_neg\_iff)`: the defect points along $-e_3 - e_4$ for $\sigma < \frac{1}{2}$ and along $+e_3 + e_4$ for $\sigma > \frac{1}{2}$ — the critical line is the unique chirality-neutral locus, the CP-odd signature of the channel;
- **$\text{SO}(8)$ orbit law** `(octTorsion\_so8\_orbit\_iff)`: FE partners’ torsions lie on a common $\text{SO}(8)$ orbit (equal norms) iff $\Re s = \frac{1}{2}$ — and since both vectors share one ray, orbit coincidence already forces vector equality `(octTorsion\_orbit\_eq\_iff\_collapse)`.

Zeros on the critical line — scope of the equivalence. `RH\_iff\_zero\_torsion\_cancellation`: $\text{RH} \Leftrightarrow$ at every nontrivial zero the FE torsion defect vanishes for every triple. The zeros are exactly the points where the $\text{SO}(8)$ twist closes the torsion loop. Because the S^7 direction is constant, cancellation is governed by the radial weight alone, and the radial weight is σ -driven — so the twist sharpens the geometry (single S^7 point, odd defect, chirality flip, orbit equivalence) without changing the logical status: “torsion cancels at every zero” is an exact RH reformulation with no extra conditions. (Frozen-table caveat: the witness direction $-e_3 - e_4$ is specific to the current octonion tables; see `AGENTS/OCTONION_TABLE_AUDIT_TODO.md`.)

8.13. The log edge: phases, Diophantine independence, and the Goldbach circle. Every equivalent condition so far has been a modulus formula. [Hqiv/Story/S3LogPhaseEdge.lean](#) formalizes why that matters and what lives on the other side.

Polar decoupling. The carrier factorizes exactly `(so4SpectralLine\_polar, log\_edge\_decoupling)`:

$$n^{-s} = n^{-\sigma} \cdot e^{-it \log n}, \quad \|e^{-it \log n}\| = 1.$$

The modulus channel knows only σ ; the unimodular phase channel knows only t . This is the machine-checked reason the modulus program produced RH-equivalences and nothing more: *no modulus formula can import phase data into a σ -constraint*. The log edge itself is `linePhase(pq) = linePhase(p) · linePhase(q)` (`linePhase\_mul`) — multiplication becomes addition in phase.

Diophantine phase independence. For distinct primes $p \neq q$, the relation $k \log p = m \log q$ (naturals or integers) forces $k = m = 0$ (`prime\_log\_nat\_rel`, `prime\_log\_int\_rel`), proved from unique factorization ($p^k = q^m$ is impossible). Consequently **two prime phases pin the height** (`two\_prime\_phases\_pin\_height`): if the phases of two distinct primes agree at heights t_1, t_2 , then $t_1 = t_2$. One prime leaves a $2\pi/\log p$ lattice of ghost heights; two incommensurable primes leave none.

The Goldbach circle: additive curvature. The additive channel supplies literal curvature (`goldbach\_pair\_circle`): a midpoint pair $p + q = 2N$ satisfies

$$pq + (q - N)^2 = N^2,$$

so Goldbach pairs are points on the circle of radius N in the (geometric-mean, deviation) plane. The multiplicative phase speed is capped by the additive constraint: $\log(pq) \leq 2 \log N$, with equality exactly at the diagonal $p = q = N$ (`pair\_phase\_speed\_max`).

Zeros are addressed by two prime phases. Combining the channels (`RH\_zero\_determined\_by\_two\_prime\_phases`): under RH, a nontrivial zero is *completely determined* by the phases of any two distinct primes at its height — σ from RH, t from phase independence. This is the precise form of the “transformer that locates a zero”: feed it two prime phases, receive the zero.

The frontier, named. The curvature is real, but it lives entirely in the t /additive channel, and decoupling cleanly separates it from σ . Any genuine forcing attack must couple the two channels — which is what the classical explicit formula does analytically through the zero set itself, and what remains open here. We name that coupling as the frontier rather than claim it.

9. EXPLICIT $h(s)$ ON THE STRIP

On the open strip with $\sigma \neq \frac{1}{2}$ the assembly collapses to the explicit quotient (Lean: [Hqiv/Story/S3InteriorStripHClosedForm.lean](#)):

$$(2) \quad h(s) = \frac{\zeta(s)}{\text{so4CriticalFactor}(s)} = \frac{\zeta(s) \sqrt{2}}{2\sigma - 1}.$$

Substituting the functional-equation assembly (1) before the quotient gives the full closed form with $\Gamma(1-s)$ and $\cos(\pi(1-s)/2)$ in the numerator (`interiorStripH\_fe\_closed\_form`): the apparent regional singularities are then *visible in h itself*, not hidden inside ζ alone.

Theorem 9.1 (Reflection symmetry of the critical factor). *For all $s \in \mathbb{C}$,*

$$\text{so4CriticalFactor}(1-s) = -\text{so4CriticalFactor}(s),$$

because the readout depends only on σ and swaps as $\sigma \leftrightarrow 1-\sigma$ (`so4CriticalFactor\_one\_sub`).

Proposition 9.2 (Reflection product seed (pathway A)). *For $0 < \sigma < 1$, $\sigma \neq \frac{1}{2}$,*

$$h(s) h(1-s) = -\frac{\zeta(s) \zeta(1-s)}{\text{so4CriticalFactor}(s)^2}.$$

Multiplying h by the equator factor recovers ζ on the strip (`completedInteriorFromH\_eq\_zeta\_on\_strip`); the main conditional predicate `InteriorStripHNonvanishingCapstone` is **logically equivalent** to `RiemannHypothesis` (`interior\_capstone\_iff\_RiemannHypothesis`)—not a weaker statement.

10. INTERIOR FACTORIZATION AND RH STATUS

On $0 < \Re s < 1$, $\sigma \neq \frac{1}{2}$, define

$$h(s) := \text{interiorStripH}(s) = \frac{\zeta(s)}{\text{so4CriticalFactor}(s)}.$$

Then $\zeta(s) = h(s) \cdot (2\sigma - 1)/\sqrt{2}$ off the line ([Hqiv/Story/S3S04InteriorWitness.lean](#)).

Definition 10.1 (Main RH conditional hypothesis). *InteriorAssemblyNonzeroAtNontrivialZerosOffLine* asserts: if $\zeta(s) = 0$ for a nontrivial zero with $\sigma \neq \frac{1}{2}$, then $h(s) \neq 0$.

Theorem 10.2 (Conditional RH). *If this hypothesis holds for interiorStripH, then RiemannHypothesis follows (RiemannHypothesis_of_SO4_interior_witness).*

Remark 10.3 (Honesty). *The factorization identity is proved; this hypothesis is not. Global factorization on the entire strip including $\sigma = \frac{1}{2}$ is false because $\zeta(\frac{1}{2} + it)$ is generally nonzero while the critical factor vanishes. The quotient witness $h = \zeta/\mathrm{so4CriticalFactor}$ proves algebra on off-line points only; it does not by itself prove the hypothesis.*

The even channel `evenStripChannel` is presently degenerate on the strip because the odd FE channel already carries the full ζ ; the physical even content is the harmonic- Δ multiplier $1 + \alpha/3$ and the $\pi^4/3$ sphere orbit (`Hqiv/Story/S3HarmonicDeltaEvenOrbit.lean`). A non-degenerate even/odd split is the remaining analytic packaging target.

11. VAPORIZATION, BACKPROJECTION, AND THE THERMODYNAMIC ARROW

The de Bruijn–Newman picture (Rodgers–Tao: $\Lambda \geq 0$, hence $\mathrm{RH} \Leftrightarrow \Lambda = 0$) reads as a heat flow that *vaporizes* off-line zeros forward in time; RH holds exactly when the present-day function sits at the vaporization front, i.e. there is **no backprojection margin** ($\Lambda \leq 0$). `Hqiv/Story/S3HeatFlowArrowNoBackprojection.lean` pollinates this picture with the already-proved HQIV temperature ladder and thermodynamics.

11.1. Dilation slot: where τ lives in $\mathfrak{so}(4, 2)$. The conformal closure of the $\mathrm{SO}(4)$ carrier is $\mathfrak{so}(4, 2) = \mathfrak{so}(4) \oplus D \oplus P_4 \oplus K_4$ with dimension count $15 = 6 + 4 + 4 + 1$ (`conformal\_dilation\_slot\_count`); the chain $\mathfrak{so}(4) \subset \mathfrak{so}(4, 2) \subset \mathfrak{so}(8)$ is $6 \leq 15 \leq 28$ (`so4\_so42\_so8\_dim\_chain`). The heat-flow deformation time τ occupies the *one-dimensional dilation slot* D ; backprojection is negative dilation. This is the same carrier escalation that produced the 45° equator readout, now read in the time channel.

11.2. Backprojection dichotomy (proved). On the discrete ladder weight $w_\tau(m) = \exp(-\tau t_{\mathrm{HQIV}}(m))$:

- **forward** ($\tau \geq 0$): $w_\tau(m) \leq 1$ on every shell and the deformed shell sum stays summable on $\Re s > 1$ (`Hqiv/Physics/HQIVHeatFlowDeformation.lean`);
- **backward** ($\tau < 0$): $w_\tau(m) > 1$ off the horizon shell (`backprojection\_weight\_gt\_one`), grows strictly shell by shell (`backprojection\_weight\_strict\_growth`), and is *unbounded* along the ladder (`backprojection\_weight\_unbounded`): no constant dominates a backward semigroup on the discrete carrier.

The proven second law (entropy-production positivity, `Hqiv/Physics/ThermodynamicLawsFromLadder.lean`) makes the arrow-induced deformation parameter nonnegative, so the physical flow can only move *toward* vaporization (`arrow\_forbids\_backprojection`, `arrow\_weight\_contractive`).

11.3. λ -lock reduction and the honest frontier. The ladder bundle carries $\lambda_{\mathrm{HQIV}} \geq 0$ — the analogue of the Rodgers–Tao inequality. Exactly as $\Lambda \geq 0$ reduces RH from $\Lambda = 0$ to the one-sided bound $\Lambda \leq 0$, the analogue zero-lock is proved equivalent to the no-backprojection inequality:

$$\lambda_{\mathrm{HQIV}} = 0 \iff \lambda_{\mathrm{HQIV}} \leq 0$$

(`lambdaHQIV\_zero\_iff\_no\_backprojection`). Concrete finite-window ladder witnesses already discharge $\lambda_{\mathrm{HQIV}} = 0$ (`lambdaHQIV\_eq\_zero\_of\_finiteWindowConcrete`).

The genuine analytic frontier is the *identification* of the discrete ladder flow with the classical ξ heat flow. It is named as the payload `VaporizationForcesCriticalLine` and proved **equivalent to RH** (`vaporization\_iff\_RiemannHypothesis`) — the same honesty pattern as the Weil-positivity

slot. Bundling the proved λ -lock side with the payload gives the `HeatFlowVaporizationBridge`; given any concrete ladder witness,

$$\text{HeatFlowVaporizationBridge inhabited} \iff \text{RiemannHypothesis}$$

(`vaporizationBridge\_iff\_RiemannHypothesis`): the thermodynamic side is free, and the vaporization payload is the entire remaining content.

11.4. Classical target formalized: Φ , H_t , and lattice inheritance. [Hqiv/Story/S3DBNLatticeInheritance.lean](#) makes the classical side concrete in Lean, so the inheritance claim is a precise statement rather than prose:

- **The density.** The de Bruijn–Newman density $\Phi(u) = \sum_{m \geq 1} (2\pi^2 m^4 e^{9u} - 3\pi m^2 e^{5u}) e^{-\pi m^2 e^{4u}}$ is `dbnPhi`; for $u \geq 0$ every term is *proved* nonnegative (`dbnPhiTerm\_nonneg`; the quartic shell dominates as soon as $2\pi m^2 e^{4u} \geq 3$, which holds on every shell since $\pi > 3$) and the series is *proved* summable (`dbnPhi\_summable`; polynomial-times-geometric domination with ratio $e^{-\pi e^{4u}} < 1$).
- **The deformed family.** $H_t(z) = \int_{u>0} e^{tu^2} \Phi(u) \cos(zu) du$ is `dbnHeatFamily`; H_0 is the (normalized) ξ .
- **Weight-level dictionary (proved).** The classical Gaussian factor *is* the repo’s discrete heat weight under $(\tau, u) \mapsto (-t, u^2)$: `dbnGaussianFactor t u = discreteHeatKernelWeight (-t) (u^2)` (`dbnGaussianFactor\_eq\_discreteHeatKernelWeight`). The dichotomy transfers: $t \leq 0$ contracts every mode (`dbn\_backward\_factor\_le\_one`), $t > 0$ amplifies without bound (`dbn\_forward\_factor\_unbounded`) — the continuum mirror of the lattice backprojection blow-up.

The bridge `DBNLatticeInheritance` then carries the classical constant Λ with the two classical theorems as explicit fields (Rodgers–Tao $\Lambda \geq 0$; Newman/de Bruijn RH $\Leftrightarrow \Lambda \leq 0$ — neither yet in Mathlib) plus the single new claim $\Lambda \leq \lambda_{\text{HQIV}}$. Since the ladder witness proves $\lambda_{\text{HQIV}} = 0$, the bridge forces $\Lambda = 0$ (`Lambda\_eq\_zero\_of\_inheritance`) and hence RH (`RiemannHypothesis\_of\_dbnLatticeInheritance`); it also instantiates the vaporization bridge (`vaporizationBridge\_of\_dbnLatticeInheritance`).

Remark 11.1 (The conviction, faithfully named). *Given the two classical imports, the lattice-domination bound is proved equivalent to RH:*

$$\Lambda \leq \lambda_{\text{HQIV}}(W) \iff \text{RiemannHypothesis}$$

(`lattice\_dominates\_iff\_RiemannHypothesis`). *The thermodynamic side is free and the classical constants are pinned; the entire content of “the discrete no-backprojection lock transfers to the continuum” is the RH frontier itself — now a single named inequality between two Lean terms rather than an informal analogy.*

11.5. Discrete dominates continuum: comparison inequality and exact phase lock. [Hqiv/Story/S3DBNDiscreteContinuumComparison.lean](#) pushes the proved boundary flush against the inheritance frontier with a genuine one-sided comparison between the two heat families, on the region where both converge absolutely. All of the following is unconditional (no hypotheses beyond parameter ranges).

Analytic control of the classical side. The dBN density is *strictly* positive for $u \geq 0$ (`dbnPhi\_pos`), decays super-exponentially, $\Phi(u) \leq C e^{-3u}$ via $e^{-\pi m^2 e^{4u}} \leq e^{-\pi(m^2-1)} e^{-\pi e^{4u}}$ and $\pi > 3$ (`dbnPhi\_le\_decay`), is continuous on $u \geq 0$ by a Weierstrass M-test (`dbnPhi\_continuousOn`), and is Lebesgue-integrable on $(0, \infty)$ (`dbnPhi\_integrableOn`). Consequently the *backward* family is uniformly bounded on the real axis:

$$t \leq 0, z \in \mathbb{R} \implies \|H_t(z)\| \leq \int_{u>0} \Phi(u) du$$

(`dbnHeatFamily\_norm\_le`), while the forward family is unbounded (`dbn\_forward\_factor\_unbounded`). At the anchor $z = 0$ the backward family is a strictly positive real (`dbnWeightedIntegral\_pos`), so its phase is exactly 0 (`dbnHeatFamily\_arg\_at\_zero`). The heat flow fixes the horizon shell. Since $t_{\text{HQIV}}(0) = 0$, the $m = 0$ weight of the discrete deformed sum is $e^{-\tau \cdot 0} = 1$ for *every* deformation time τ (`hqivHeatKernelWeight\_horizon`). At trivial twist ($\varphi = 0$) and real anchor $s = \sigma > 1$ the full complex sum is the coercion of a convergent positive real series (`HQIVDeformedSum\_real\_anchor`), so the flow-invariant horizon term gives the unconditional lower bound $\text{eff}_0^{-\sigma} \leq \|\text{HQIVDeformedSum}\|$ (`HQIVDeformedSum\_anchor\_lower`): the flow can never dim the horizon, and every other shell only adds positive mass.

The comparison inequality and phase alignment. Combining the two sides, for every $\tau \geq 0$, $t \leq 0$, $\sigma > 1$ and every real z there is an *explicit* constant $c = \text{eff}_0^{-\sigma} / (\int \Phi + 1) > 0$ with

$$c \|H_t(z)\| \leq \|\text{HQIVDeformedSum } \tau \dots \sigma\|$$

(`discrete\_dominates\_continuum\_anchor`): no backward continuum deformation can outgrow the lattice sum — the growth sign is preserved. The phase-alignment statement holds in its strongest form: the arguments of the discrete sum and the backward continuum family *coincide* at the anchor, so the cosine of their difference is exactly 1 (`phase\_alignment\_anchor`), in particular bounded below by $\frac{1}{2}$ (`phase\_alignment\_cos\_ge\_half`).

Remark 11.2 (Honest scope). *The comparison lives on the absolutely-convergent region ($\Re s > 1$ discrete $\times$ real axis continuum, all $t \leq 0$). Extending it into the critical strip with zero tracking is precisely the inheritance bound $\Lambda \leq \lambda_{\text{HQIV}}$, which `lattice\_dominates\_iff\_RiemannHypothesis` shows is RH itself. The module advances the proved boundary up to that frontier; it does not cross it.*

11.6. The constant itself: Λ ported as a Lean definition. `Hqiv/Story/S3DBNConstantPort.lean` removes the last abstraction from the inheritance bridge: the classical constant is no longer an arbitrary real packaged in a structure field but a *concrete Lean term*,

$$\text{dbnLambda} = \inf \{ t \in \mathbb{R} \mid \forall z, H_t(z) = 0 \Rightarrow \Im z = 0 \},$$

the infimum of the hyperbolicity locus of the very `dbnHeatFamily` whose Gaussian factor was proved identical to the lattice heat kernel. The two classical imports are now precisely typed statements about *that* constant — the form required for an actual Mathlib contribution:

- `RodgersTaoStatement` := $0 \leq \text{dbnLambda}$ (Rodgers–Tao, Annals 191 (2020));
- `NewmanDBNStatement` := $\text{RiemannHypothesis} \Leftrightarrow \text{dbnLambda} \leq 0$ (Newman 1976; de Bruijn 1950);
- `DBNUpwardClosure` — de Bruijn’s heat-flow preservation of hyperbolicity (the locus is an upper ray): the continuum analytic mirror of the lattice no-backprojection dichotomy;
- `DBNXiBridge` — the normalization $H_0(z) \propto \xi(\frac{1}{2} + \frac{iz}{2})$ against Mathlib’s `completedRiemannZeta`, stated normalization-robustly.

Unconditionally proved on top of the definition: the backward family is nondegenerate at the anchor, $H_t(0) \neq 0$ for $t \leq 0$ (`dbnHeatFamily\_ne\_zero\_at\_anchor`); the inf interface (`dbnLambda\_le\_of\_realZerosAt`, `le\_dbnLambda\_of\_forall`); upper-set propagation under de Bruijn closure (`dbnRealZerosAt\_mono\_of\_upwardClosure`); and, given the two classical statements, $\text{dbnLambda} = 0 \Leftrightarrow \text{RH}$ (`dbnLambda\_eq\_zero\_iff\_RH`). The honesty theorems of the inheritance bridge now hold verbatim at the concrete constant:

$$\text{dbnLambda} \leq \lambda_{\text{HQIV}}(W) \iff \text{RiemannHypothesis}$$

(`concrete\_lattice\_dominates\_iff\_RiemannHypothesis`, `concrete\_dbnLatticeInheritance\_iff\_RiemannHypothesis`). Porting the *proofs* of Rodgers–Tao and Newman/de Bruijn (Laguerre–Pólya theory, heat-flow zero dynamics) remains a major formalization programme; the objects and statements now have their precise Lean targets.

12. THE SHARED $\frac{1}{2}$ BRIDGE: RIEMANN HYPOTHESIS AND GOLDBACH

12.1. Two classical problems, one projection constant. The Riemann Hypothesis and Goldbach's conjecture (even parity case: every even $n > 2$ is a sum of two primes) are unrelated in classical analytic number theory. In the HQIV $\mathfrak{so}(8)$ / $\mathrm{SO}(4)$ projection story they meet on the same normalized constant $\frac{1}{2}$:

Channel	Geometric object	$\frac{1}{2}$ meaning
ζ / zeros	critical line $\Re s = \frac{1}{2}$	$\text{rot45Free}(\sigma) = 0 \Leftrightarrow \sigma = \frac{1}{2}$
Prime sums	midpoint N , pair (p, q) with $p + q = 2N$	tangent slope $N/(p + q) = \frac{1}{2}$

The ζ row is the pointwise equator condition of §3. The Goldbach row is proved *for any certified midpoint pair*, without assuming Goldbach:

$$\frac{N}{p+q} = \frac{N}{2N} = \frac{1}{2} \quad \text{whenever } p + q = 2N$$

([Hqiv/Geometry/GoldbachG2Parity.lean](#), `so4\_orthogonal\_tangent\_midpoint\_slope\_eq\_half`). Diagonal partitions $N + N = 2N$ when N is prime carry the same slope; there is no exception branch (`so4\_orthogonal\_diagonal\_tangent\_slope\_eq\_half`).

12.2. Midpoint Goldbach and the holonomy bridge. Lean names the arithmetic target in *midpoint* form: for each midpoint N , primes $p \leq N \leq q$ with $p + q = 2N$ (`GoldbachMidpointPair`). Eventual midpoint Goldbach above a threshold N_0 is `MidpointGoldbachEventually  $N_0$` ; even Goldbach parity (`GoldbachParity`) follows at $N_0 = 2$ via `goldbach_pair_of_midpoint_pair`.

The open arithmetic–geometric bridge is positivity of the finite paired-prime count around each midpoint:

$$\text{goldbachMidpointCount}(N) > 0 \implies \exists p, q : \text{GoldbachMidpointPair } N \, p \, q,$$

packaged as `S04ZetaHolonomyForcesMidpointPairs  $N_0$` (`midpoint\_goldbach\_of\_so4\_zeta\_holonomy\_bridge`). Any certified midpoint pair lifts to integrable Hopf-shell holonomy with Δ/G_2 /triality fields (`hopf\_fiber\_midpoint\_holonomy\_support\_of\_midpoint\_pair`).

The constructive $\mathrm{G}_2 + \Delta$ harmonic landing layer ([Hqiv/Geometry/GoldbachG2Parity.lean](#)) states the complementary finite certificate target `DeltaHarmonicCompleteness`: once every even $n > 2$ has a Δ -harmonic landing, `goldbach_from_delta_harmonic` extracts `GoldbachParity` with no extra arithmetic input.

12.3. The bridge is $\mathrm{RH} \wedge \text{Goldbach}$: proved equivalence. [Hqiv/Story/S3ExplicitFormulaDualitySlot.lean](#) names the shared payload explicitly. An `S08ProjectedHalfSlopeBridge  $N_0$` bundles:

- **ζ side.** `WeilPositivityForcesCriticalLine`, i.e. the explicit-formula / positivity localization step equivalent to RH (`weilPositivity\_iff\_RiemannHypothesis`);
- **prime side.** `S04ZetaHolonomyForcesMidpointPairs  $N_0$` .

Both directions are now machine-checked, so the packaging is a proved *logical equivalence*, not merely a sufficient condition:

(3)

$$\text{S08ProjectedHalfSlopeBridge } N_0 \iff \text{RiemannHypothesis} \wedge \text{MidpointGoldbachEventually } N_0,$$

(4)

$$\text{S08ProjectedHalfSlopeBridge } 2 \iff \text{RiemannHypothesis} \wedge \text{GoldbachParity}.$$

Equation (3) is `so4\_projected\_half\_slope\_iff\_rh\_and\_midpoint\_goldbach`; equation (4) is the capstone `so4\_projected\_half\_slope\_two\_iff\_rh\_and\_goldbach\_parity`. The converse direction is built from two reconstruction lemmas:

- a certified midpoint pair forces a positive midpoint count (`midpoint\_count\_pos\_of\_midpoint\_pair`), so the holonomy payload is *equivalent* to eventual midpoint Goldbach (`so4\_zeta\_holonomy\_bridge\_iff\_midpoint\_goldbach`);
- an unordered Goldbach pair for $2N$ reorders into a midpoint pair at N (`midpoint\_pair\_of\_goldbach\_pair\_two\_mul`), giving `MidpointGoldbachEventually 2` $\Leftrightarrow$ `GoldbachParity (midpoint\_goldbach\_two\_iff\_goldbach\_parity)`.

The channels are also separated: the ζ field alone is exactly RH, and the midpoint field at threshold 2 alone is exactly Goldbach parity, with no leakage between them (`bridge\_channels\_are\_rh\_and\_goldbach, so4\_zeta\_holonomy\_bridge\_two\_iff\_goldbach\_parity`).

Remark 12.1 (What “RH = Goldbach” means here). *This note does not claim a classical theorem $\text{RH} \Leftrightarrow \text{Goldbach}$ in Mathlib. What is proved is the identification of the geometric payload with the conjunction: the half-slope bridge is **provably the same proposition** as $\text{RH} \wedge \text{Goldbach parity}$ (equation (4)), with the same normalized $\frac{1}{2}$ readout serving as the critical-line lock on the ζ channel and the midpoint tangent slope on the prime channel. The packaging therefore has zero slack: discharging the bridge is exactly as hard as the two open problems combined—by the equivalence theorem, provably so—and conversely any future proof of RH and Goldbach immediately instantiates the bridge. Neither classical conjecture is proved here; what is discharged is the faithfulness of the geometric encoding itself.*

12.4. Prime side of the explicit formula. The von Mangoldt weight $\Lambda(n) = \log p$ on prime powers is the concrete prime-power content dual to the zero sum in the explicit formula ([Hqiv/Story/S3ExplicitFormulaDualitySlot.lean](#), `chebyshevPsi`, `vonMangoldt\_prime\_eq\_log`). Positivity on that prime side is the standard route to critical-line zero localization; in this repo it is named faithfully as RH-equivalent, parallel to the interior capstone on $h(s)$.

13. PTE DEGREE 3 AND $\text{SO}(4)$ CANCELLATION

Recent work of Lee, Takahashi, and Tsai [6] shows that anomaly cancellation in a chiral hidden sector with unbroken vector-like $U(1)_H \times U(1)_X$ is *equivalent* to the degree $k = 3$ Prouhet–Tarry–Escott (PTE) problem: two equal-size multisets $A = \{a_i\}$, $B = \{b_i\}$ must satisfy

$$\sum_i a_i^\ell = \sum_i b_i^\ell \quad (\ell = 1, 2, 3).$$

Minimal ideal solutions have size $n = k + 1 = 4$, matching the four-dimensional $\text{SO}(4)$ quaternion carrier of §2. The phenomenological prediction—at least four hidden mass eigenstates, typically organized as near-degenerate doublets when the PTE solution is sign-symmetric—parallels the six-pole geometry of §4: antipodal pairs $(+i, -i)$, $(+j, -j)$, $(+k, -k)$ cancel as *orbits* while each pole retains a nonzero $\pm$ residual ([Hqiv/Story/S3SixPolesResidual.lean](#)).

13.1. Sign-symmetric multisets and head/tail reflection. Lee *et al.* focus on **symmetric** PTE solutions $A = \{\pm a_1, \dots, \pm a_m\}$, $B = \{\pm b_1, \dots, \pm b_m\}$. Under this sign pairing, the $\ell = 1$ and $\ell = 3$ constraints are automatic (both sides sum to zero), so the only nontrivial ideal- $n = 4$ condition is

$$a_1^2 + a_2^2 = b_1^2 + b_2^2,$$

a matching of *even* (quadratic) invariants. This is the same parity split as head/tail reflection on the S^3 shell:

$$\text{criticalProj}(p) + \text{criticalProj}(\text{headTailReflect } p) = 0$$

(`headTail\_orbit\_pair\_cancels`) forces **linear** ($\ell = 1$) cancellation for every orbit pair, while single-axis poles never vanish pointwise (`criticalProj\_ne\_zero\_of\_singleAxis`). Geometrically, the PTE even constraint is orthogonal invariance of a two-vector norm; the 45° readout in §3 is the same rotation that fixes the diagonal slot at $1/\sqrt{2}$ and isolates the equator slot $(2\sigma - 1)/\sqrt{2}$.

13.2. Orbit vs. pointwise: two readings of “cancellation”. Lee *et al.* distinguish PTE solutions related by affine charge shifts (number-theoretically equivalent, physically distinct mass/coupling assignments) from the orbit-level doublet structure their symmetric solutions predict. The S^3 story makes the same distinction internally:

- **Orbit cancellation** — reflection pairs $\{\sigma, 1 - \sigma\}$ and head/tail pairs always sum to zero in the free coordinate ([Hqiv/Story/S3OrbitVsPointwiseGap.lean](#), `orbit\_free\_sum\_cancels`); this holds even off $\Re s = \frac{1}{2}$.
- **Pointwise cancellation** — a *single* slot vanishes iff $\sigma = \frac{1}{2}$ (Theorem 3.2); upgrading orbit cancellation to pointwise cancellation at actual ζ zeros is the RH capstone (§10).

Reading PTE through this lens: matching power sums is the discrete charge-theoretic avatar of orbit cancellation; selecting a *physical* spectrum (mass doublets, minicharge partners) is the pointwise upgrade—an analogue of RH-hard localization on the critical line.

13.3. Extension toward hidden-sector model building. Three concrete bridges from this note to [6]:

- (i) **Lean packaging.** Formalize symmetric PTE- $k = 3$ solutions as charge multisets with proved automatic $\ell = 1, 3$ cancellation under sign pairing, parallel to [Hqiv/Story/S3PrimeAxisCancellation.lean](#).
- (ii) **Ideal $n = 4$ sum-of-squares.** Embed minimal PTE solutions as integer samples on the $\pm i, \pm j$ (or $\pm j, \pm k$) pole lattice; the surviving constraint $a_1^2 + a_2^2 = b_1^2 + b_2^2$ is then a two-axis norm match on S^3 .
- (iii) **TUFT hidden readout.** In the HQIV patch ontology, finite SM anomaly traces are proved at the coefficient level ([Hqiv/Algebra/AnomalyCancellation.lean](#)); extending the same finite-sum discipline to $U(1)_H \times U(1)_X$ content would import Lee *et al.*’s PTE equivalence as a *model-building corollary* of the $SO(4)$ cancellation geometry, not as an external axiom.

14. PATHWAYS TOWARD AN UNCONDITIONAL PROOF

The analysis below matches the claim-status table: the capstone is RH-hard. Plausible extensions that exploit *this* framework (not imported from external regularization claims) include:

- A. Completed h .** Use (2) and `so4CriticalFactor(1 - s) = - so4CriticalFactor(s)` to build a candidate entire/completed interior function; check whether reflection forces off-line zeros to contradict discharge lemmas.
- B. Zero-free re-expression.** Rewrite h as a manifestly nonzero Dirichlet/Euler or harmonic-weighted integral on $\sigma \neq \frac{1}{2}$.
- C. $SO(4)$ /patch obstruction.** Relate an off-line zero to a nonzero holonomy or commutator on the octonion carrier, contradicting proved `so(8)` closure.
- D. Classical tools in channel language.** Phragmén–Lindelöf on vertical lines $\sigma \neq \frac{1}{2}$ using the $\tan(\pi s/2)$ ratio; explicit-formula with separated sin/cos slots.
- E. Non-degenerate even split.** Promote the harmonic- Δ multiplier into `evenStripChannel` so h is a genuine even/odd assembly rather than a pure quotient.
- F. Shared half-slope bridge (§12).** Prove `S04ZetaHolonomyForcesMidpointPairs` from finite patch holonomy on the same $\frac{1}{2}$ readout that locks ζ -zeros to $\Re s = \frac{1}{2}$, or prove `DeltaHarmonicCompleteness` on the $G_2 + \Delta$ landing channel.
- G. PTE hidden-sector packaging (§13).** Import [6] symmetric PTE- $k = 3$ solutions as charge samples on the six-pole lattice; use sum-of-squares matching as the even-channel counterpart to `rot45Diag` rigidity.

15. LEAN MODULE INDEX

Build target: `lake build HQIVStory`. Core modules (import order in [HQIVStory.lean](#)):

- [Hqiv/Story/S3ClosureDeltaLiftBridge.lean](#) — harmonic $H_n \leq K(n)$, $SO(4)$ lift;

- [Hqiv/Story/S3HarmonicPrimeZetaPath.lean](#) — harmonic $\Rightarrow$ primes $\Rightarrow \zeta \Rightarrow \Lambda$;
- [Hqiv/Story/S3FortyFiveProjection.lean](#) — 45° diagonal / equator;
- [Hqiv/Story/S3RotationRigidity.lean](#) — projection line, twiddle rigidity;
- [Hqiv/Story/S3MirroredTwiddlePair.lean](#) — mirrored twiddle pairs at θ and $\pi/2 - \theta$, symmetric zero loci, FE orbit cancellation, critical-line coincidence;
- [Hqiv/Story/S3CubicPlasticTwiddle.lean](#) — higher-twiddle factorization probes: linear free_c, plastic so4PlasticCubicFree, mirror pairs; not a zero-locator channel (§3.3);
- [Hqiv/Story/S3HigherTwiddleFactorizationProbe.lean](#) — per-channel twiddleInteriorAssembly, patch factorization off probe loci, semiprime-supported finite cross-talk, frame-radius obstruction;
- [Hqiv/Story/S3ZetaClosedForm.lean](#) — regional closed forms, factorization predicate;
- [Hqiv/Story/S3S04ZetaProjectionClosedForm.lean](#) — $\pi^4/6$ sphere slot, master packaging;
- [Hqiv/Story/S3HarmonicDeltaEvenOrbit.lean](#) — $1 + \alpha/3 = 6/5$ orbit;
- [Hqiv/Story/S3ZetaAxisRotationProjection.lean](#) — j/k cancellation and ratios;
- [Hqiv/Story/S3S04InteriorWitness.lean](#) — even/odd channels, interiorStripH;
- [Hqiv/Story/S3FESlotDischarge.lean](#) — negative-integer slot discharge;
- [Hqiv/Story/S3DeltaOrbitOffStrip.lean](#) — off-strip confinement;
- [Hqiv/Story/S3SigmaReadoutScope.lean](#) — σ -blindness, no global $\zeta = \text{readout}$;
- [Hqiv/Story/S3ZeroQuadrupletOrbit.lean](#) — FE pairs, Schwarz quadruplet;
- [Hqiv/Story/S3OrbitVsPointwiseGap.lean](#) — orbit vs pointwise RH gap;
- [Hqiv/Story/S3RHDischarge.lean](#) — discharge packaging;
- [Hqiv/Story/S3InteriorStripHClosedForm.lean](#) — explicit $h(s)$, reflection symmetry, capstone $\Leftrightarrow$ RH;
- [Hqiv/Story/S3ExplicitFormulaDualitySlot.lean](#) — prime side $\psi(x)$, S08ProjectedHalfSlopeBridge, RH $\wedge$ Goldbach joint packaging;
- [Hqiv/Geometry/GoldbachG2Parity.lean](#) — midpoint slope $\frac{1}{2}$, holonomy bridge, $G_2 + \Delta$ landing certificates;
- [Hqiv/Story/S3HeatFlowArrowNoBackprojection.lean](#) — dilation slot count, backprojection dichotomy, arrow $\Rightarrow$ no backprojection, λ -lock, HeatFlowVaporizationBridge $\Leftrightarrow$ RH;
- [Hqiv/Story/S3DBNLatticeInheritance.lean](#) — concrete Φ and H_t , term positivity and summability, weight-level lattice dictionary, DBNLatticeInheritance, inheritance bound $\Leftrightarrow$ RH;
- [Hqiv/Story/S3DBNDiscreteContinuumComparison.lean](#) — strict positivity, decay, continuity and integrability of Φ ; uniform bound on backward H_t ; flow-invariant horizon term; comparison inequality $c\|H_t(z)\| \leq \|\text{HQIVDeformedSum}\|$ and exact phase alignment;
- [Hqiv/Story/S3DBNConstantPort.lean](#) — the de Bruijn–Newman constant dbnLambda as a concrete inf, Rodgers–Tao and Newman/dBN as precisely typed statements, honesty theorems at the concrete constant;
- [Hqiv/Story/S3TangentOrbitCriticalLine.lean](#) — tangent law of the quadruplet orbit: FE inverts, Schwarz conjugates, modulus identity $\|\sin\|^2 - \|\cos\|^2 = -\cos(\pi\sigma)$, critical line = unimodular locus, orbit collapse $\Leftrightarrow$ line, equator factor tie;
- [Hqiv/Story/S3EulerSpectralCancellation.lean](#) — zeros as spectral cancellations: non-vanishing Euler factors and no cancellation in the product region, breakdown-locus confinement, tame von Mangoldt spectrum, resonance blow-up of $\|1/\zeta\|$ at every zero, spectral-cancellation profile with the tangent orbit attached;
- [Hqiv/Story/S3SpectralResonanceChanneling.lean](#) — collective cancellation (single prime lines and finite truncations never vanish for $\Re s > 0$), two-channel dichotomy at every zero, geometric-channel absorption $\Leftrightarrow$ RH, $\text{SO}(4)$ spectral-line transformer with FE

pairing, geometric-mean and square-root-weight locators, prime generation of lines, $\text{RH} \Leftrightarrow$ square-root spectral weights;

- [Hqiv/Story/S3VerticalAxisRigidity.lean](#) — vertical mirror and its fixed line, FE closure of the zero set, two-mirrors-make-a-translation rigidity (abstract and zeta-instantiated unique axis), FE-partner collinearity pin, square-root-weight locus as a globally straight vertical line with flow invariance;
- [Hqiv/Story/S3SameHeightOrbitCollapse.lean](#) — σ -uniform collectivity of spectral weights, same-height quadruplet partner and merge law, $\text{RH} \Leftrightarrow$ height pins $\mathfrak{R}$ (FE only), $\text{RH} \Leftrightarrow$ one zero per height (with Schwarz hypothesis);
- [Hqiv/Story/S3PolarProjectionCollapse.lean](#) — polar partner and same-height law, polar separation = equator factor ($-\sqrt{2} \cdot \text{so4CriticalFactor}$), separation norm = twice the line deviation, collapse law, single-point polar projection $\Leftrightarrow \text{RH}$, chain to one-zero-per-height;
- [Hqiv/Story/S3ZeroHolonomyGoldbachChain.lean](#) — pair activation at every zero, AM-GM joint-weight cap $1/(pq) \geq 1/N^2$, zero-contains-pair-holonomy capstone, parity gives holonomy support, bridge $\Leftrightarrow$ spectral weights $\wedge$ parity $\Leftrightarrow$ polar collapse $\wedge$ parity;
- [Hqiv/Story/S3TuftNestedFrameTower.lean](#) — level- N spectral frame radius, nesting recursion, σ -blind flow rigidity, positivity, harmonic radius law at every level $N \geq 2$, $\text{RH} \Leftrightarrow$ harmonic tower radii at every zero, divergence of the ladder on the line, curvature domination $H_N \leq K(N)$;
- [Hqiv/Story/S3DiophantineTransformer.lean](#) — level- N diagonal transformer $D_N(s)$, trace ladder $\rightarrow \zeta$ on $\mathfrak{R}s > 1$, HS norm = frame radius, FE composition = fixed rational harmonic operator $\text{diag}(1/n)$ with trace H_N , adjoint law $D_N(1-s) = D_N(s)^\dagger \Leftrightarrow$ line, scaled isometry on the line, $\text{RH} \Leftrightarrow$ adjoint law at every zero;
- [Hqiv/Story/S3OctonionicAssociatorChannel.lean](#) — commutative saturation of the transformer family (abelian, normal), trilinear associator scaling, concrete witness $[e_1, e_2, e_4] = -e_3 - e_4$ with $\text{norm}^2 = 2$, associator channel $2(abc)^{-2\sigma}$, positivity at every zero, square-root locator, FE pairing $4/(abc)^2$, $\text{RH} \Leftrightarrow$ torsion weights at zeros, Goldbach floor $1/N^3$ on the line;
- [Hqiv/Story/S3OctonionS7TorsionCancellation.lean](#) — torsion vector $= (abc)^{-\sigma} \cdot (-e_3 - e_4)$ (single S^7 point), FE defect closed form and oddness, cancellation $\Leftrightarrow$ line, chirality flip across the line, $\text{SO}(8)$ orbit law and orbit $\Rightarrow$ vector collapse, $\text{RH} \Leftrightarrow$ torsion cancellation at every zero;
- [Hqiv/Story/S3CriticalLineCarrierBundle.lean](#) — line/off-line carrier contrast: maximal Hopf circle and unimodular tangent only on $\mathfrak{R}s = \frac{1}{2}$, shrunk fiber and hyperbolic tangent sorting off-line, Path C via rolling on the line;
- [Hqiv/Story/S3HarmonicShellZeroCounting.lean](#) — height t as non-injective cover; harmonic-shell slots (n, k) at arc angles $2\pi k/n$; reciprocal j - k chart at $k/(nH_n)$; sweep refines under $n \mid m$; countable slot encoding; balance / conditional ζ -zero probes;
- [Hqiv/Story/S3AnalyticStripLift.lean](#) — cylinder lift `stripPointLift`, 45° free coordinate and Hopf radius, 2π periodicity (`sameStripHeight`), σ -equator $= \frac{1}{2}$, whole-strip twiddle;
- [Hqiv/Story/S3AnalyticStripClassification.lean](#) — conditional zero classification on the full strip (cancellation, Path C/E, rolling recovery on the line, centered model);
- [Hqiv/Story/S3LogPhaseEdge.lean](#) — polar decomposition and decoupling theorem, log edge $\text{linePhase}(pq) = \text{linePhase}(p)\text{linePhase}(q)$, prime-log incommensurability over $\mathbb{N}$ and $\mathbb{Z}$, two-prime phase height pinning, Goldbach circle $pq + (q - N)^2 = N^2$, phase-speed cap with diagonal equality case, $\text{RH} \Rightarrow$ zeros addressed by two prime phases.

APPENDIX A. CLAIM STATUS TABLE

APPENDIX B. NOTATION

α	HQIV imprint 3/5 (<code>alpha_eq_3_5</code>)
H_n	harmonic partial sum (<code>harmonicPartialSum</code>)
(n, k)	harmonic-shell slot, angle $2\pi k/n$ (<code>HarmonicShellSlot</code>)
$K(n)$	curvature integral (<code>curvature_integral</code>)
<code>so4CriticalFactor(s)</code>	$(2\sigma - 1)/\sqrt{2}$
<code>oddStripChannel</code>	FE assembly (<code>1</code>)
<code>interiorStripH</code>	$(\text{even} + \text{odd})/\text{so4CriticalFactor}$
S, C	$\sin(\pi s/2), \cos(\pi s/2)$

REFERENCES

1. Steven Jr Ettinger, *Dynamic fermion mass spectrum and standard model lagrangian from the HQIV discrete octonion action: Nielsen TUFT hopf shells and inner-outer casimir realization*, Zenodo (HQIV community), 2026, Version v2 (2026-06-08). Tier-2 #5 (combined opener). Unifies Nielsen/TUFT Hopf-fibration necessity, the discrete octonion-carrier realization, the dynamic inner-outer Casimir mechanism with explicit κ_6 factorisation, and the sector-by-sector projection of the proved discrete octonion action onto $\mathcal{L}_{\text{SM}}$. Includes `scripts.zip`. Source: `papers/tuft.sm.lagrangian/`. Lean target: `paper.tuft.sm.lagrangian`.
2. ———, *Heavy-flavour decay chains from HQIV three-ledger readout: Discrete multichannel rules, quarkonium contact, and branching without partial-width inputs*, Preprint, HQIV-LEAN repository (`papers/hep.decay.readout/`), 2026, Tier-2 decay-calculator extension; cites `hqiv-tuft-sm-lagrangian-paper` (DOI 10.5281/zenodo.20601215) and `hqiv-nucleon-binding-paper`. HEP mass panel, programmatic multichannel expansion ($\sim 190 J/\psi$, $\sim 222 \Upsilon$ channels), hidden-quarkonium EM contact 39/10, strict benchmark discharge (87/87).
3. ———, *Light-cone-derived auxiliary fields supply an explicit derived interpretation of quantum superposition*, Zenodo (HQIV community), 2026, Concept DOI; v2 DOI 10.5281/zenodo.19336553 (2026-03-30). Octonionic $\mathbb{R}^8$ carrier, lossless QM bookkeeping, $\alpha = 3/5$ / $\gamma = 2/5$ forced from the lattice.
4. ———, *Review: From discrete null-lattice growth to $so(8)$: Concrete realization and exact lie closure*, Zenodo (HQIV community), 2026, Concept DOI; latest version DOI 10.5281/zenodo.20214211 (2026-05-15). Includes `closure.pdf` and `so8_closure_full_appendix.pdf`.
5. Ettinger, Steven Jr. and HQIV contributors, *Hqiv lean 4 formalization (slimmed public mirror)*, GitHub (HQIV organization), 2026, Public, slimmed-down Lean 4 development carrying the modules cited from the HQIV paper series. Companion Zenodo manuscript DOI 10.5281/zenodo.18899939.
6. Junseok Lee, Fuminobu Takahashi, and Yu-Dai Tsai, *Number theory in quantum physics: Minicharged particles and the Prouhet-Tarry-Escott problem*, (2026), TU-1300; anomaly cancellation for vector-like $U(1)_{\text{H}} \times U(1)_{\text{X}}$ hidden sectors equivalent to PTE degree $k = 3$.

INDEPENDENT RESEARCHER

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Claim	Lean	RH-hard
$H_n \leq K(n)$, K strictly increasing	yes	no
$\langle \mathfrak{so}(3), \Delta_4 \rangle = \mathfrak{so}(4)$	yes	no
Shell sum $= \zeta(s)$ for $\Re s > 1$	yes	no
Prime Euler product $= \zeta(s)$	yes	no
FE strip closed form (1)	yes	no
Bernoulli / even- π regional values	yes	no
Even $s = 2k$: $\mathcal{S}(s) = 0$, $\mathcal{C}(s) = (-1)^k$	yes	no
Odd $s = 2k + 1$: $\mathcal{C}(s) = 0$	yes	no
$1 + \alpha/3 = 6/5$ harmonic-third projection	yes	no
Nontrivial zeros $\notin$ negative integers	yes	no
Nontrivial zeros in open strip $0 < \Re s < 1$	yes	no
$\zeta = h \cdot (2\sigma - 1)/\sqrt{2}$ off $\sigma = \frac{1}{2}$	yes	no
Readout factors through $\Re s$ only	yes	no
No global $\zeta = \text{exactTwiddleReadout}$	yes	no
FE pair at nontrivial zeros	yes	no
Quadruplet zeros (with $\zeta(\bar{s}) = \overline{\zeta(s)}$)	yes	no
Midpoint tangent slope $N/(p+q) = \frac{1}{2}$ for pairs	yes	no
Hopf holonomy support for midpoint pairs	yes	no
Bridge payload $\Leftrightarrow$ midpoint Goldbach	yes	no
Bridge at 2 $\Leftrightarrow \text{GoldbachParity}$	yes	no
$\text{S08ProjectedHalfSlopeBridge2} \Leftrightarrow \text{RH} \wedge \text{Goldbach}$	yes	no
$\mathfrak{so}(4, 2)$ dilation slot count $15 = 6 + 4 + 4 + 1$	yes	no
Backward weights > 1 , strict growth, unbounded	yes	no
Arrow (2nd law) $\Rightarrow$ no backprojection	yes	no
$\lambda_{\text{HQIV}} = 0 \Leftrightarrow \lambda_{\text{HQIV}} \leq 0$	yes	no
Vaporization bridge $\Leftrightarrow \text{RH}$ (given witness)	yes	no
Φ terms ≥ 0 and summable ($u \geq 0$)	yes	no
Gaussian factor = lattice weight at $(-t, u^2)$	yes	no
$\Phi > 0$, continuous, $\Phi \leq Ce^{-3u}$, integrable on $(0, \infty)$	yes	no
$\ H_t(z)\ \leq \int \Phi$ for $t \leq 0$, z real	yes	no
Horizon shell weight = 1 for all τ	yes	no
$c\ H_t(z)\ \leq \ \text{HQIVDeformedSum}\ $ (anchor region)	yes	no
Phase alignment $\cos(\Delta \arg) = 1$ at anchor	yes	no
$\text{dbnLambda} = \inf$ hyperbolicity locus (concrete def.)	yes	no
$H_t(0) \neq 0$ for $t \leq 0$ (anchor nondegeneracy)	yes	no
Tangent orbit law $T(1-s) = T(s)^{-1}$, $T(\bar{s}) = \overline{T(s)}$	yes	no
$\ \sin\ ^2 - \ \cos\ ^2 = -\cos(\pi\sigma)$ at $\pi s/2$	yes	no
Critical line $\Leftrightarrow \ \tan(\pi s/2)\ = 1$ (strip)	yes	no
Strip cylinder lift on S^3 ; $t \sim t + 2\pi n$ same point	yes	no
Line carrier: $f = 0$, $r = 1$, unimodular T , orbit collapse	yes	no
Off-line: $r < 1$, tangent sorts half-planes, no quadruplet collapse	yes	no
Whole-strip cancellation $f(\sigma) + r(\sigma)(\cos t + \sin t) = 0$	yes	yes
Path C/E classification under lift + bridge	yes	yes
Tangent orbit collapse $T^{-1} = \bar{T} \Leftrightarrow \text{line}$	yes	no
Equator factor = 0 $\Leftrightarrow$ unimodular tangent	yes	no
Euler factors nonvanishing; no cancellation for $\Re s > 1$	yes	no
Zero $\Rightarrow$ product breakdown ($\Re s < 1$, incl. boundary)	yes	no
Tame spectrum: $\sum \Lambda(n)n^{-s} = -\zeta'/\zeta$ for $\Re s > 1$	yes	no
Resonance: $\ 1/\zeta\ \rightarrow \infty$ at every zero $\rho \neq 1$	yes	no
Spectral-cancellation profile of nontrivial zeros	yes	no
Single prime lines/finite truncations $\neq 0$ for $\Re s > 0$	yes	no
Two-channel dichotomy at every nontrivial zero	yes	no
$\text{InteriorAssemblyNonzero} \Leftrightarrow \text{RH}$ (equivalence)	yes	no
FE pairing $\ n^{-s}\ \ n^{-(1-s)}\ = 1/n$ (all s)	yes	no
Locators: geometric mean / $\ n^{-s}\ ^2 = 1/n \Leftrightarrow \text{line}$	yes	no
$\text{RH} \Leftrightarrow$ square-root spectral weights at zeros	yes	no
Mirror fixed set = axis line; FE closure of zero set	yes	no